



Standard Practice for Conversion of Kinematic Viscosity to Saybolt Universal Viscosity or to Saybolt Furol Viscosity¹

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This standard has been approved for use by agencies of the Department of Defense.

1. Scope*

1.1 This practice² covers the conversion tables and equations for converting kinematic viscosity in mm^2/s at any temperature to Saybolt Universal viscosity in Saybolt Universal seconds (SUS) at the same temperature and for converting kinematic viscosity in mm^2/s at 122 and 210°F (50 and 98.9°C) to Saybolt Furol viscosity in Saybolt Furol seconds (SFS) at the same temperatures. Kinematic viscosity values are based on water being 1.0034 mm^2/s (cSt) at 68°F (20°C).

NOTE 1—A fundamental and preferred method for measuring kinematic viscosity is by use of kinematic viscometers as outlined in Test Method D 445. It is recommended that kinematic viscosity be reported in millimetres squared per second, instead of Saybolt Universal Seconds (SUS) or Saybolt Furol Seconds (SFS). This method is being retained for the purpose of calculation of kinematic viscosities from SUS and SFS data that appear in past literature. One millimetre squared per second (mm^2/s) equals one centistoke (cSt), which is another unit commonly found in older literature.

1.2 The SI unit of kinematic viscosity, mm^2/s , and temperature in degrees Fahrenheit are the standard in this practice.

2. Referenced Documents

2.1 ASTM Standards:³

D 445 Test Method for Kinematic Viscosity of Transparent and Opaque Liquids

¹ This practice is under the jurisdiction of ASTM Committee D02 on Petroleum Products and Lubricants and is the direct responsibility of Subcommittee D02.07 on Flow Properties.

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² This practice, together with Practice D 2270, replaces Compilation of ASTM Viscosity Tables for Kinematic Viscosity Conversions.

³ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

D 2270 Practice for Calculating Viscosity Index from Kinematic Viscosity at 40 and 100°C

2.2 ASTM Adjunct:

ADJD2161 Viscosity Extrapolation Tables to Zero Degrees Fahrenheit (SSU)

3. Summary of Practice

3.1 The Saybolt Universal viscosity equivalent to a given kinematic viscosity varies with the temperature at which the determination is made. The basic conversion values are those given in Table 1 for 100°F. The Saybolt Universal viscosity equivalent to a given kinematic viscosity at any temperature may be calculated as described in 4.3. Equivalent values at 210°F are given in Table 1 for convenience.

3.2 The Saybolt Furol viscosity equivalents are tabulated in Table 3 for temperatures of 122°F and 210°F only.

3.3 Examples for using the tables are given in Appendix X1.

4. Significance and Use

4.1 At one time the petroleum industry relied on measuring kinematic viscosity by means of the Saybolt viscometer, and expressing kinematic viscosity in units of Saybolt Universal Seconds (SUS) and Saybolt Furol Seconds (SFS). This practice is now obsolete in the petroleum industry.

4.2 This practice establishes the official equations relating SUS and SFS to the SI kinematic viscosity units, mm^2/s .

4.3 This practice allows for the conversion between SUS and SFS units and SI units of kinematic viscosity.

5. Procedure for Conversion to Saybolt Universal Viscosity

5.1 Convert kinematic viscosities between 1.81 and 500 mm^2/s (cSt) at 100°F, and between 1.77 and 139.8 mm^2/s (cSt) at 210°F, to equivalent Saybolt Universal seconds directly from Table 1 (see Appendix X1, Example 1).

NOTE 2—Obtain viscosities not listed, but which are within the range given in Table 1, by linear interpolation (see Appendix X1, Example 2).

*A Summary of Changes section appears at the end of this standard.

5.2 Convert kinematic viscosities greater than the upper limits of **Table 1** at temperatures of 100 and 210°F to Saybolt Universal viscosities as follows (see **Appendix X1**, Example 3):

$$\text{Saybolt Universal seconds} = \text{centistokes} \times B \quad (1)$$

where $B = 4.632$ at 100°F or 4.664 at 210°F.

5.3 At temperatures other than 100 or 210°F, convert kinematic viscosities to Saybolt Universal viscosities as follows⁴ (see **Appendix X1**, Example 4):

$$U_t = U_{100^\circ\text{F}}(1 + 0.000061(t - 100)) \quad (2)$$

where:

U_t = Saybolt Universal viscosity at $t^\circ\text{F}$, and
 $U_{100^\circ\text{F}}$ = Saybolt Universal viscosity at 100°F in Saybolt Universal seconds equivalent to kinematic viscosity in centistokes at $t^\circ\text{F}$, from **Table 1**.

NOTE 3—The multipliers for Saybolt Universal seconds in Eq 2 are given as Factor A in **Table 2** for a range of temperatures.

5.4 Since the relationship between Saybolt and kinematic viscosities is linear above 75 mm²/s (cSt), kinematic viscosities above this limit may be converted to Saybolt Universal viscosities at any temperature between 0 and 350°F by use of Eq 1 (4.2), selecting the proper factor for B from **Table 2** (see **Appendix X1**, Example 5).

6. Procedure for Conversion to Saybolt Furol Viscosity

6.1 Convert kinematic viscosities between 48 to 1300 mm²/s (cSt) at 122°F, and between 50 and 1300 mm²/s (cSt) at 210°F, to equivalent Saybolt Furol seconds directly from **Table 3** (see **Appendix X1**, Examples 6 and 7).

NOTE 4—Viscosities not listed, but which are within the range given in **Table 3**, may be obtained by linear interpolation (see **Appendix X1**, Example 8).

6.2 Convert kinematic viscosities above 1300 cSt to equivalent Saybolt Furol seconds by use of the following equations (see **Appendix X1**, Example 9):

$$\begin{aligned} \text{Saybolt Furol seconds at } 122^\circ\text{F} \\ = 0.4717 \times \text{mm}^2/\text{s (cSt) at } 122^\circ\text{F} \end{aligned} \quad (3)$$

$$\begin{aligned} \text{Saybolt Furol seconds at } 210^\circ\text{F} \\ = 0.4792 \times \text{mm}^2/\text{s (cSt) at } 210^\circ\text{F} \end{aligned} \quad (4)$$

7. Procedure for Computer Calculation

7.1 **Table 1** and **Table 3** were computed by fitting a smooth curve to the original experimental data points. The derived equations are given as follows for the convenience of those who wish to use a computer for conversion rather than refer to the tables:

$$U_{100^\circ\text{F}} = 4.6324\nu + \frac{1.0 + 0.03264\nu}{(3930.2 + 262.7\nu + 23.97\nu^2 + 1.646\nu^3) \times 10^{-5}} \quad (5)$$

$$U_t = [1.0 + 0.000061(t - 100)] \cdot \left[4.6324\nu + \frac{1.0 + 0.03264\nu}{(3930.2 + 262.7\nu + 23.97\nu^2 + 1.646\nu^3) \times 10^{-5}} \right] \quad (6)$$

$$F_{122^\circ\text{F}} = 0.4717\nu + \left[\frac{13924}{(\nu^2 - 72.59\nu + 6816)} \right] \quad (7)$$

$$F_{210^\circ\text{F}} = 0.4792\nu + \left[\frac{5610}{(\nu^2 + 2130)} \right] \quad (8)$$

where:

ν = kinematic viscosity, mm²/s (cSt) at $t^\circ\text{F}$,
 $F_{122^\circ\text{F}}$ = Saybolt Furol viscosity at 122°F in Saybolt Furol seconds equivalent to kinematic viscosity, mm²/s (cSt) at 122°F, and
 $F_{210^\circ\text{F}}$ = Saybolt Furol viscosity at 210°F in Saybolt Furol seconds equivalent to kinematic viscosity, mm²/s (cSt) at 210°F.

7.2 Eq 5 and Eq 6 and **Table 1** are limited to values of Saybolt Universal of 32.0 s and above.

7.3 Eq 7 and Eq 8 and **Table 3** are limited to values of Saybolt Furol of 25.1 s and above.

8. Supplementary Conversion Equivalents

8.1 The following units and equivalents are frequently used in connection with viscosity conversions:

poise	= cgs unit of absolute viscosity.
centipoise	= 0.01 poise.
stokes	= cgs unit of kinematic viscosity.
centistokes	= 0.01 stokes.
centipoise	= centistokes $\times$ density (at temperature under consideration).

9. Report

9.1 Saybolt Universal and Saybolt Furol viscosities should be reported to the nearest 0.1 s for values below 200 s and to the nearest whole second for values of 200 s and higher.

10. Keywords

10.1 kinematic viscosity; Saybolt furol; Saybolt universal

⁴ "Viscosity Extrapolation Tables to Zero Degrees Fahrenheit (SSU)" are available at a nominal cost from ASTM International Headquarters. Request Adjunct No. ADJD2161.

**TABLE 1 Kinematic Viscosity to Saybolt Universal Viscosity
1.77 to 500.0 mm²/s (cSt)**

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
			2.25	33.5	33.7	2.75	35.1	35.4	3.25	36.8	37.0
			2.26	33.5	33.7	2.76	35.2	35.4	3.26	36.8	37.0
1.77	..	32.0	2.27	33.5	33.7	2.77	35.2	35.4	3.27	36.8	37.1
1.78	..	32.1	2.28	33.6	33.8	2.78	35.2	35.5	3.28	36.9	37.1
1.79	..	32.1	2.29	33.6	33.8	2.79	35.3	35.5	3.29	36.9	37.1
1.80	..	32.1	2.30	33.6	33.8	2.80	35.3	35.5	3.30	36.9	37.2
1.81	32.0	32.2	2.31	33.7	33.9	2.81	35.3	35.6	3.31	37.0	37.2
1.82	32.0	32.2	2.32	33.7	33.9	2.81	35.4	35.6	3.32	37.0	37.2
1.83	32.0	32.2	2.33	33.7	33.9	2.83	35.4	35.6	3.33	37.0	37.3
1.84	32.1	32.3	2.34	33.8	34.0	2.84	35.4	35.7	3.34	37.1	37.3
1.85	32.1	32.3	2.35	33.8	34.0	2.85	35.5	35.7	3.35	37.1	37.3
1.86	32.1	32.3	2.36	33.8	34.0	2.86	35.5	35.7	3.36	37.1	37.4
1.87	32.2	32.4	2.37	33.9	34.1	2.87	35.5	35.8	3.37	37.2	37.4
1.88	32.2	32.4	2.38	33.9	34.1	2.88	35.6	35.8	3.38	37.2	37.4
1.89	32.2	32.4	2.39	33.9	34.2	2.89	35.6	35.8	3.39	37.2	37.5
1.90	32.3	32.5	2.40	34.0	34.2	2.90	35.6	35.9	3.40	37.3	37.5
1.91	32.3	32.5	2.41	34.0	34.2	2.91	35.7	35.9	3.41	37.3	37.5
1.92	32.3	32.5	2.42	34.0	34.3	2.92	35.7	35.9	3.42	37.3	37.6
1.93	32.4	32.6	2.43	34.1	34.3	2.93	35.7	36.0	3.43	37.4	37.6
1.94	32.4	32.6	2.44	34.1	34.3	2.94	35.8	36.0	3.44	37.4	37.6
1.95	32.4	32.6	2.45	34.1	34.4	2.95	35.8	36.0	3.45	37.4	37.7
1.96	32.5	32.7	2.46	34.2	34.4	2.96	35.8	36.1	3.46	37.5	37.7
1.97	32.5	32.7	2.47	34.2	34.4	2.97	35.9	36.1	3.47	37.5	37.7
1.98	32.5	32.8	2.48	34.2	34.5	2.98	35.9	36.1	3.48	37.5	37.8
1.99	32.6	32.8	2.49	34.3	34.5	2.99	35.9	36.2	3.49	37.6	37.8
2.00	32.6	32.8	2.50	34.3	34.5	3.00	36.0	36.2	3.50	37.6	37.8
2.01	32.6	32.9	2.51	34.3	34.6	3.01	36.0	36.2	3.51	37.6	37.9
2.02	32.7	32.9	2.52	34.4	34.6	3.02	36.0	36.3	3.52	37.6	37.9
2.03	32.7	32.9	2.53	34.4	34.6	3.03	36.0	36.3	3.53	37.7	37.9
2.04	32.7	33.0	2.54	34.4	34.7	3.04	36.1	36.3	3.54	37.7	38.0
2.05	32.8	33.0	2.55	34.5	34.7	3.05	36.1	36.4	3.55	37.7	38.0
2.06	32.8	33.0	2.56	34.5	34.7	3.06	36.1	36.4	3.56	37.8	38.0
2.07	32.8	33.1	2.57	34.5	34.8	3.07	36.2	36.4	3.57	37.8	38.1
2.08	32.9	33.1	2.58	34.6	34.8	3.08	36.2	36.5	3.58	37.8	38.1
2.09	32.9	33.1	2.59	34.6	34.8	3.09	36.2	36.5	3.59	37.9	38.1
2.10	32.9	33.2	2.60	34.6	34.9	3.10	36.3	36.5	3.60	37.9	38.2
2.11	33.0	33.2	2.61	34.7	34.9	3.11	36.3	36.6	3.61	37.9	38.2
2.12	33.0	33.2	2.62	34.7	34.9	3.12	36.3	36.6	3.62	38.0	38.2
2.13	33.0	33.3	2.63	34.7	35.0	3.13	36.4	36.6	3.63	38.0	38.3
2.14	33.1	33.3	2.64	34.8	35.0	3.14	36.4	36.7	3.64	38.0	38.3
2.15	33.1	33.3	2.65	34.8	35.0	3.15	36.4	36.7	3.65	38.1	38.3
2.16	33.1	33.4	2.66	34.8	35.1	3.16	36.5	36.7	3.66	38.1	38.4
2.17	33.2	33.4	2.67	34.9	35.1	3.17	36.5	36.8	3.67	38.1	38.4
2.18	33.2	33.4	2.68	34.9	35.1	3.18	36.5	36.8	3.68	38.2	38.4
2.19	33.2	33.5	2.69	34.9	35.2	3.19	36.6	36.8	3.69	38.2	38.5
2.20	33.3	33.5	2.70	35.0	35.2	3.20	36.6	36.9	3.70	38.2	38.5
2.21	33.3	33.5	2.71	35.0	35.2	3.21	36.6	36.9	3.71	38.3	38.5
2.22	33.3	33.6	2.72	35.0	35.3	3.22	36.7	36.9	3.72	38.3	38.6
2.23	33.4	33.6	2.73	35.1	35.3	3.23	36.7	37.0	3.73	38.3	38.6
2.24	33.4	33.6	2.74	35.1	35.3	3.24	36.7	37.0	3.74	38.4	38.6

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
3.75	38.4	38.7	4.25	40.0	40.3	4.75	41.6	41.9	5.25	43.2	43.5
3.76	38.4	38.7	4.26	40.0	40.3	4.76	41.6	41.9	5.26	43.2	43.5
3.77	38.5	38.7	4.27	40.1	40.3	4.77	41.7	41.9	5.27	43.3	43.5
3.78	38.5	38.7	4.28	40.1	40.4	4.78	41.7	42.0	5.28	43.3	43.6
3.79	38.5	38.8	4.29	40.1	40.4	4.79	41.7	42.0	5.29	43.3	43.6
3.80	38.6	38.8	4.30	40.2	40.4	4.80	41.8	42.0	5.30	43.3	43.6
3.81	38.6	38.8	4.31	40.2	40.5	4.81	41.8	42.1	5.31	43.4	43.7
3.82	38.6	38.9	4.32	40.2	40.5	4.82	41.8	42.1	5.32	43.4	43.7
3.83	38.7	38.9	4.33	40.3	40.5	4.83	41.9	42.1	5.33	43.4	43.7
3.84	38.7	38.9	4.34	40.3	40.6	4.84	41.9	42.2	5.34	43.5	43.8
3.85	38.7	39.0	4.35	40.3	40.6	4.85	41.9	42.2	5.35	43.5	43.8
3.86	38.7	39.0	4.36	40.4	40.6	4.86	41.9	42.2	5.36	43.5	43.8
3.87	38.8	39.0	4.37	40.4	40.7	4.87	42.0	42.3	5.37	43.6	43.9
3.88	38.8	39.1	4.38	40.4	40.7	4.88	42.0	42.3	5.38	43.6	43.9
3.89	38.8	39.1	4.39	40.4	40.7	4.89	42.0	42.3	5.39	43.6	43.9
3.90	38.9	39.1	4.40	40.5	40.8	4.90	42.1	42.4	5.40	43.7	44.0
3.91	38.9	39.2	4.41	40.5	40.8	4.91	42.1	42.4	5.41	43.7	44.0
3.92	38.9	39.2	4.42	40.5	40.8	4.92	42.1	42.4	5.42	43.7	44.0
3.93	39.0	39.2	4.43	40.6	40.8	4.93	42.2	42.5	5.43	43.8	44.1
3.94	39.0	39.3	4.44	40.6	40.9	4.94	42.2	42.5	5.44	43.8	44.1
3.95	39.0	39.3	4.45	40.6	40.9	4.95	42.2	42.5	5.45	43.8	44.1
3.96	39.1	39.3	4.46	40.7	40.9	4.96	42.3	42.5	5.46	43.9	44.2
3.97	39.1	39.4	4.47	40.7	41.0	4.97	42.3	42.6	5.47	43.9	44.2
3.98	39.1	39.4	4.48	40.7	41.0	4.98	42.3	42.6	5.48	43.9	44.2
3.99	39.2	39.4	4.49	40.8	41.0	4.99	42.4	42.6	5.49	44.0	44.2
4.00	39.2	39.5	4.50	40.8	41.1	5.00	42.4	42.7	5.50	44.0	44.3
4.01	39.2	39.5	4.51	40.8	41.1	5.01	42.4	42.7	5.51	44.0	44.3
4.02	39.3	39.5	4.52	40.9	41.1	5.02	42.5	42.7	5.52	44.0	44.3
4.03	39.3	39.6	4.53	40.9	41.2	5.03	42.5	42.8	5.53	44.1	44.4
4.04	39.3	39.6	4.54	40.9	41.2	5.04	42.5	42.8	5.54	44.1	44.4
4.05	39.4	39.6	4.55	41.0	41.2	5.05	42.6	42.8	5.55	44.1	44.4
4.06	39.4	39.7	4.56	41.0	41.3	5.06	42.6	42.9	5.56	44.2	44.5
4.07	39.4	39.7	4.57	41.0	41.3	5.07	42.6	42.9	5.57	44.2	44.5
4.08	39.5	39.7	4.58	41.1	41.3	5.08	42.6	42.9	5.58	44.2	44.5
4.09	39.5	39.8	4.59	41.1	41.4	5.09	42.7	43.0	5.59	44.3	44.6
4.10	39.5	39.8	4.60	41.1	41.4	5.10	42.7	43.0	5.60	44.3	44.6
4.11	39.6	39.8	4.61	41.2	41.4	5.11	42.7	43.0	5.61	44.3	44.6
4.12	39.6	39.8	4.62	41.2	41.5	5.12	42.8	43.1	5.62	44.4	44.7
4.13	39.6	39.9	4.63	41.2	41.5	5.13	42.8	43.1	5.63	44.4	44.7
4.14	39.6	39.9	4.64	41.2	41.5	5.14	42.8	43.1	5.64	44.4	44.7
4.15	39.7	39.9	4.65	41.3	41.6	5.15	42.9	43.2	5.65	44.5	44.8
4.16	39.7	40.0	4.66	41.3	41.6	5.16	42.9	43.2	5.66	44.5	44.8
4.17	39.7	40.0	4.67	41.3	41.6	5.17	42.9	43.2	5.67	44.5	44.8
4.18	39.8	40.0	4.68	41.4	41.7	5.18	43.0	43.3	5.68	44.6	44.9
4.19	39.8	40.1	4.69	41.4	41.7	5.19	43.0	43.3	5.69	44.6	44.9
4.20	39.8	40.1	4.70	41.4	41.7	5.20	43.0	43.3	5.70	44.6	44.9
4.21	39.9	40.1	4.71	41.5	41.7	5.21	43.1	43.3	5.71	44.7	45.0
4.22	39.9	40.2	4.72	41.5	41.8	5.22	43.1	43.4	5.72	44.7	45.0
4.23	39.9	40.2	4.73	41.5	41.8	5.23	43.1	43.4	5.73	44.7	45.0
4.24	40.0	40.2	4.74	41.6	41.8	5.24	43.2	43.4	5.74	44.7	45.0

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
5.75	44.8	45.1	6.25	46.4	46.7	6.75	48.0	48.3	7.25	49.6	49.9
5.76	44.8	45.1	6.26	46.4	46.7	6.76	48.0	48.3	7.26	49.6	50.0
5.77	44.8	45.1	6.27	46.4	46.8	6.77	48.0	48.4	7.27	49.7	50.0
5.78	44.9	45.2	6.28	46.5	46.8	6.78	48.1	48.4	7.28	49.7	50.0
5.79	44.9	45.2	6.29	46.5	46.8	6.79	48.1	48.4	7.29	49.7	50.1
5.80	44.9	45.2	6.30	46.5	46.8	6.80	48.1	48.5	7.30	49.8	50.1
5.81	45.0	45.3	6.31	46.6	46.9	6.81	48.2	48.5	7.31	49.8	50.1
5.82	45.0	45.3	6.32	46.6	46.9	6.82	48.2	48.5	7.32	49.8	50.2
5.83	45.0	45.3	6.33	46.6	46.9	6.83	48.2	48.6	7.33	49.9	50.2
5.84	45.1	45.4	6.34	46.7	47.0	6.84	48.3	48.6	7.34	49.9	50.2
5.85	45.1	45.4	6.35	46.7	47.0	6.85	48.3	48.6	7.35	49.9	50.3
5.86	45.1	45.4	6.36	46.7	47.0	6.86	48.3	48.7	7.36	50.0	50.3
5.87	45.2	45.5	6.37	46.8	47.1	6.87	48.4	48.7	7.37	50.0	50.3
5.88	45.2	45.5	6.38	46.8	47.1	6.88	48.4	48.7	7.38	50.0	50.4
5.89	45.2	45.5	6.39	46.8	47.1	6.89	48.4	48.8	7.39	50.1	50.4
5.90	45.3	45.6	6.40	46.9	47.2	6.90	48.5	48.8	7.40	50.1	50.4
5.91	45.3	45.6	6.41	46.9	47.2	6.91	48.5	48.8	7.41	50.1	50.5
5.92	45.3	45.6	6.42	46.9	47.2	6.92	48.5	48.9	7.42	50.2	50.5
5.93	45.4	45.7	6.43	47.0	47.3	6.93	48.6	48.9	7.43	50.2	50.5
5.94	45.4	45.7	6.44	47.0	47.3	6.94	48.6	48.9	7.44	50.2	50.6
5.95	45.4	45.7	6.45	47.0	47.3	6.95	48.6	49.0	7.45	50.3	50.6
5.96	45.4	45.8	6.46	47.0	47.4	6.96	48.7	49.0	7.46	50.3	50.6
5.97	45.5	45.8	6.47	47.1	47.4	6.97	48.7	49.0	7.47	50.3	50.7
5.98	45.5	45.8	6.48	47.1	47.4	6.98	48.7	49.1	7.48	50.3	50.7
5.99	45.5	45.9	6.49	47.1	47.5	6.99	48.8	49.1	7.49	50.4	50.7
6.00	45.6	45.9	6.50	47.2	47.5	7.00	48.8	49.1	7.50	50.4	50.8
6.01	45.6	45.9	6.51	47.2	47.5	7.01	48.8	49.1	7.51	50.4	50.8
6.02	45.6	45.9	6.52	47.2	47.6	7.02	48.9	49.2	7.52	50.5	50.8
6.03	45.7	46.0	6.53	47.3	47.6	7.03	48.9	49.2	7.53	50.5	50.9
6.04	45.7	46.0	6.54	47.3	47.6	7.04	48.9	49.2	7.54	50.5	50.9
6.05	45.7	46.0	6.55	47.3	47.7	7.05	49.0	49.3	7.55	50.6	50.9
6.06	45.8	46.1	6.56	47.4	47.7	7.06	49.0	49.3	7.56	50.6	51.0
6.07	45.8	46.1	6.57	47.4	47.7	7.07	49.0	49.3	7.57	50.6	51.0
6.08	45.8	46.1	6.58	47.4	47.8	7.08	49.0	49.4	7.58	50.7	51.0
6.09	45.9	46.2	6.59	47.5	47.8	7.09	49.1	49.4	7.59	50.7	51.0
6.10	45.9	46.2	6.60	47.5	47.8	7.10	49.1	49.4	7.60	50.7	51.1
6.11	45.9	46.2	6.61	47.5	47.8	7.11	49.1	49.5	7.61	50.8	51.1
6.12	46.0	46.3	6.62	47.6	47.9	7.12	49.2	49.5	7.62	50.8	51.1
6.13	46.0	46.3	6.63	47.6	47.9	7.13	49.2	49.5	7.63	50.8	51.2
6.14	46.0	46.3	6.64	47.6	47.9	7.14	49.2	49.6	7.64	50.9	51.2
6.15	46.1	46.4	6.65	47.7	48.0	7.15	49.3	49.6	7.65	50.9	51.2
6.16	46.1	46.4	6.66	47.7	48.0	7.16	49.3	49.6	7.66	50.9	51.3
6.17	46.1	46.4	6.67	47.7	48.0	7.17	49.3	49.7	7.67	51.0	51.3
6.18	46.2	46.5	6.68	47.8	48.1	7.18	49.4	49.7	7.68	51.0	51.3
6.19	46.2	46.5	6.69	47.8	48.1	7.19	49.4	49.7	7.69	51.0	51.4
6.20	46.2	46.5	6.70	47.8	48.1	7.20	49.4	49.8	7.70	51.1	51.4
6.21	46.2	46.6	6.71	47.9	48.2	7.21	49.5	49.8	7.71	51.1	51.4
6.22	46.3	46.6	6.72	47.9	48.2	7.22	49.5	49.8	7.72	51.1	51.5
6.23	46.3	46.6	6.73	47.9	48.2	7.23	49.5	49.9	7.73	51.2	51.5
6.24	46.3	46.7	6.74	47.9	48.3	7.24	49.6	49.9	7.74	51.2	51.5

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
7.75	51.2	51.6	8.25	52.9	53.2	8.75	54.6	54.9	9.25	56.3	56.6
7.76	51.3	51.6	8.26	52.9	53.3	8.76	54.6	55.0	9.26	56.3	56.7
7.77	51.3	51.6	8.27	53.0	53.3	8.77	54.6	55.0	9.27	56.3	56.7
7.78	51.3	51.7	8.28	53.0	53.3	8.78	54.7	55.0	9.28	56.4	56.7
7.79	51.4	51.7	8.29	53.0	53.4	8.79	54.7	55.1	9.29	56.4	56.8
7.80	51.4	51.7	8.30	53.1	53.4	8.80	54.7	55.1	9.30	56.4	56.8
7.81	51.4	51.8	8.31	53.1	53.4	8.81	54.8	55.1	9.31	56.5	56.8
7.82	51.5	51.8	8.32	53.1	53.5	8.82	54.8	55.2	9.32	56.5	56.9
7.83	51.5	51.8	8.33	53.2	53.5	8.83	54.8	55.2	9.33	56.5	56.9
7.84	51.5	51.9	8.34	53.2	53.5	8.84	54.9	55.2	9.34	56.6	56.9
7.85	51.6	51.9	8.35	53.2	53.6	8.85	54.9	55.3	9.35	56.6	57.0
7.86	51.6	51.9	8.36	53.3	53.6	8.86	54.9	55.3	9.36	56.6	57.0
7.87	51.6	52.0	8.37	53.3	53.6	8.87	55.0	55.3	9.37	56.7	57.0
7.88	51.7	52.0	8.38	53.3	53.7	8.88	55.0	55.4	9.38	56.7	57.1
7.89	51.7	52.0	8.39	53.4	53.7	8.89	55.0	55.4	9.39	56.7	57.1
7.90	51.7	52.1	8.40	53.4	53.7	8.90	55.1	55.4	9.40	56.8	57.1
7.91	51.8	52.1	8.41	53.4	53.8	8.91	55.1	55.5	9.41	56.8	57.2
7.92	51.8	52.1	8.42	53.5	53.8	8.92	55.1	55.5	9.42	56.8	57.2
7.93	51.8	52.2	8.43	53.5	53.8	8.93	55.2	55.5	9.43	56.9	57.2
7.94	51.9	52.2	8.44	53.5	53.9	8.94	55.2	55.6	9.44	56.9	57.3
7.95	51.9	52.2	8.45	53.6	53.9	8.95	55.2	55.6	9.45	56.9	57.3
7.96	51.9	52.3	8.46	53.6	53.9	8.96	55.3	55.6	9.46	57.0	57.4
7.97	52.0	52.3	8.47	53.6	54.0	8.97	55.3	55.7	9.47	57.0	57.4
7.98	52.0	52.3	8.48	53.7	54.0	8.98	55.3	55.7	9.48	57.0	57.4
7.99	52.0	52.4	8.49	53.7	54.0	8.99	55.4	55.7	9.49	57.1	57.5
8.00	52.1	52.4	8.50	53.7	54.1	9.00	55.4	55.8	9.50	57.1	57.5
8.01	52.1	52.4	8.51	53.8	54.1	9.01	55.4	55.8	9.52	57.2	57.6
8.02	52.1	52.5	8.52	53.8	54.1	9.02	55.5	55.8	9.54	57.2	57.6
8.03	52.2	52.5	8.53	53.8	54.2	9.03	55.5	55.9	9.56	57.3	57.7
8.04	52.2	52.5	8.54	53.9	54.2	9.04	55.5	55.9	9.58	57.4	57.8
8.05	52.2	52.6	8.55	53.9	54.2	9.05	55.6	55.9	9.60	57.5	57.8
8.06	52.3	52.6	8.56	53.9	54.3	9.06	55.6	56.0	9.62	57.5	57.9
8.07	52.3	52.6	8.57	54.0	54.3	9.07	55.6	56.0	9.64	57.6	58.0
8.08	52.3	52.7	8.58	54.0	54.3	9.08	55.7	56.0	9.66	57.7	58.0
8.09	52.4	52.7	8.59	54.0	54.4	9.09	55.7	56.1	9.68	57.7	58.1
8.10	52.4	52.7	8.60	54.1	54.4	9.10	55.7	56.1	9.70	57.8	58.2
8.11	52.4	52.8	8.61	54.1	54.5	9.11	55.8	56.1	9.72	57.9	58.3
8.12	52.5	52.8	8.62	54.1	54.5	9.12	55.8	56.2	9.74	57.9	58.3
8.13	52.5	52.8	8.63	54.2	54.5	9.13	55.8	56.2	9.76	58.0	58.4
8.14	52.5	52.9	8.64	54.2	54.6	9.14	55.9	56.3	9.78	58.1	58.5
8.15	52.6	52.9	8.65	54.2	54.6	9.15	55.9	56.3	9.80	58.1	58.5
8.16	52.6	52.9	8.66	54.3	54.6	9.16	55.9	56.3	9.82	58.2	58.6
8.17	52.6	53.0	8.67	54.3	54.7	9.17	56.0	56.4	9.84	58.3	58.7
8.18	52.7	53.0	8.68	54.3	54.7	9.18	56.0	56.4	9.86	58.4	58.7
8.19	52.7	53.0	8.69	54.4	54.7	9.19	56.0	56.4	9.88	58.4	58.8
8.20	52.7	53.1	8.70	54.4	54.8	9.20	56.1	56.5	9.90	58.5	58.9
8.21	52.8	53.1	8.71	54.4	54.8	9.21	56.1	56.5	9.92	58.6	59.0
8.22	52.8	53.1	8.72	54.5	54.8	9.22	56.2	56.5	9.94	58.6	59.0
8.23	52.8	53.2	8.73	54.5	54.9	9.23	56.2	56.6	9.96	58.7	59.1
8.24	52.9	53.2	8.74	54.5	54.9	9.24	56.2	56.6	9.98	58.8	59.2

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
10.00	58.8	59.2	11.00	62.4	62.8	12.00	66.0	66.4	13.00	69.7	70.2
10.02	58.9	59.3	11.02	62.4	62.9	12.02	66.1	66.5	13.02	69.8	70.3
10.04	59.0	59.4	11.04	62.5	62.9	12.04	66.1	66.6	13.04	69.9	70.3
10.06	59.0	59.4	11.06	62.6	63.0	12.06	66.2	66.7	13.06	69.9	70.4
10.08	59.1	59.5	11.08	62.7	63.1	12.08	66.3	66.7	13.08	70.0	70.5
10.10	59.2	59.6	11.10	62.7	63.1	12.10	66.4	66.8	13.10	70.1	70.6
10.12	59.3	59.7	11.12	62.8	63.2	12.12	66.4	66.9	13.12	70.2	70.6
10.14	59.3	59.7	11.14	62.9	63.3	12.14	66.5	67.0	13.14	70.2	70.7
10.16	59.4	59.8	11.16	62.9	63.4	12.16	66.6	67.0	13.16	70.3	70.8
10.18	59.5	59.9	11.18	63.0	63.4	12.18	66.7	67.1	13.18	70.4	70.9
10.20	59.5	59.9	11.20	63.1	63.5	12.20	66.7	67.2	13.20	70.5	70.9
10.22	59.6	60.0	11.22	63.2	63.6	12.22	66.8	67.2	13.22	70.5	71.0
10.24	59.7	60.1	11.24	63.2	63.7	12.24	66.9	67.3	13.24	70.6	71.1
10.26	59.7	60.1	11.26	63.3	63.7	12.26	66.9	67.4	13.26	70.7	71.2
10.28	59.8	60.2	11.28	63.4	63.8	12.28	67.0	67.5	13.28	70.8	71.2
10.30	59.9	60.3	11.30	63.4	63.9	12.30	67.1	67.5	13.30	70.8	71.3
10.32	60.0	60.4	11.32	63.5	63.9	12.32	67.2	67.6	13.32	70.9	71.4
10.34	60.0	60.4	11.34	63.6	64.0	12.34	67.2	67.7	13.34	71.0	71.5
10.36	60.1	60.5	11.36	63.7	64.1	12.36	67.3	67.8	13.36	71.1	71.5
10.38	60.2	60.6	11.38	63.7	64.2	12.38	67.4	67.8	13.38	71.1	71.6
10.40	60.2	60.6	11.40	63.8	64.2	12.40	67.5	67.9	13.40	71.2	71.7
10.42	60.3	60.7	11.42	63.9	64.3	12.42	67.5	68.0	13.42	71.3	71.8
10.44	60.4	60.8	11.44	63.9	64.4	12.44	67.6	68.1	13.44	71.4	71.9
10.46	60.4	60.9	11.46	64.0	64.5	12.46	67.7	68.1	13.46	71.4	71.9
10.48	60.5	60.9	11.48	64.1	64.5	12.48	67.8	68.2	13.48	71.5	72.0
10.50	60.6	61.0	11.50	64.2	64.6	12.50	67.8	68.3	13.50	71.6	72.1
10.52	60.7	61.1	11.52	64.2	64.7	12.52	67.9	68.4	13.52	71.7	72.2
10.54	60.7	61.1	11.54	64.3	64.7	12.54	68.0	68.4	13.54	71.8	72.2
10.56	60.8	61.2	11.56	64.4	64.8	12.56	68.1	68.5	13.56	71.8	72.3
10.58	60.9	61.3	11.58	64.5	64.9	12.58	68.1	68.6	13.58	71.9	72.4
10.60	60.9	61.4	11.60	64.5	65.0	12.60	68.2	68.7	13.60	72.0	72.5
10.62	61.0	61.4	11.62	64.6	65.0	12.62	68.3	68.7	13.62	72.1	72.5
10.64	61.1	61.5	11.64	64.7	65.1	12.64	68.4	68.8	13.64	72.1	72.6
10.66	61.2	61.6	11.66	64.7	65.2	12.66	68.4	68.9	13.66	72.2	72.7
10.68	61.2	61.6	11.68	64.8	65.3	12.68	68.5	69.0	13.68	72.3	72.8
10.70	61.3	61.7	11.70	64.9	65.3	12.70	68.6	69.0	13.70	72.4	72.8
10.72	61.4	61.8	11.72	65.0	65.4	12.72	68.7	69.1	13.72	72.4	72.9
10.74	61.4	61.9	11.74	65.0	65.5	12.74	68.7	69.2	13.74	72.5	73.0
10.76	61.5	61.9	11.76	65.1	65.5	12.76	68.8	69.3	13.76	72.6	73.1
10.78	61.6	62.0	11.78	65.2	65.6	12.78	68.9	69.3	13.78	72.7	73.2
10.80	61.7	62.1	11.80	65.3	65.7	12.80	69.0	69.4	13.80	72.7	73.2
10.82	61.7	62.1	11.82	65.3	65.8	12.82	69.0	69.5	13.82	72.8	73.3
10.84	61.8	62.2	11.84	65.4	65.8	12.84	69.1	69.6	13.84	72.9	73.4
10.86	61.9	62.3	11.86	65.5	65.9	12.86	69.2	69.6	13.86	73.0	73.5
10.88	61.9	62.4	11.88	65.6	66.0	12.88	69.3	69.7	13.88	73.1	73.5
10.90	62.0	62.4	11.90	65.6	66.1	12.90	69.3	69.8	13.90	73.1	73.6
10.92	62.1	62.5	11.92	65.7	66.1	12.92	69.4	69.9	13.92	73.2	73.7
10.94	62.2	62.6	11.94	65.8	66.2	12.94	69.5	69.9	13.94	73.3	73.8
10.96	62.2	62.6	11.96	65.8	66.3	12.96	69.6	70.0	13.96	73.4	73.9
10.98	62.3	62.7	11.98	65.9	66.4	12.98	69.6	70.1	13.98	73.4	73.9

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
14.00	73.5	74.0	15.00	77.4	77.9	16.00	81.4	81.9	17.00	85.4	86.0
14.02	73.6	74.1	15.02	77.5	78.0	16.02	81.4	82.0	17.02	85.5	86.0
14.04	73.7	74.2	15.04	77.6	78.1	16.04	81.5	82.1	17.04	85.6	86.1
14.06	73.7	74.2	15.06	77.6	78.2	16.06	81.6	82.2	17.06	85.6	86.2
14.08	73.8	74.3	15.08	77.7	78.2	16.08	81.7	82.2	17.08	85.7	86.3
14.10	73.9	74.4	15.10	77.8	78.3	16.10	81.8	82.3	17.10	85.8	86.4
14.12	74.0	74.5	15.12	77.9	78.4	16.12	81.8	82.4	17.12	85.9	86.5
14.14	74.1	74.6	15.14	78.0	78.5	16.14	81.9	82.5	17.14	86.0	86.5
14.16	74.1	74.6	15.16	78.0	78.6	16.16	82.0	82.6	17.16	86.0	86.6
14.18	74.2	74.7	15.18	78.1	78.6	16.18	82.1	82.6	17.18	86.1	86.7
14.20	74.3	74.8	15.20	78.2	78.7	16.20	82.2	82.7	17.20	86.2	86.8
14.22	74.4	74.9	15.22	78.3	78.8	16.22	82.2	82.8	17.22	86.3	86.9
14.24	74.4	74.9	15.24	78.3	78.9	16.24	82.3	82.9	17.24	86.4	86.9
14.26	74.5	75.0	15.26	78.4	79.0	16.26	82.4	83.0	17.26	86.5	87.0
14.28	74.6	75.1	15.28	78.5	79.0	16.28	82.5	83.0	17.28	86.5	87.1
14.30	74.7	75.2	15.30	78.6	79.1	16.30	82.6	83.1	17.30	86.6	87.2
14.32	74.7	75.3	15.32	78.7	79.2	16.32	82.6	83.2	17.32	86.7	87.3
14.34	74.8	75.3	15.34	78.7	79.3	16.34	82.7	83.3	17.34	86.8	87.4
14.36	74.9	75.4	15.36	78.8	79.3	16.36	82.8	83.4	17.36	86.9	87.4
14.38	75.0	75.5	15.38	78.9	79.4	16.38	82.9	83.4	17.38	86.9	87.5
14.40	75.1	75.6	15.40	79.0	79.5	16.40	83.0	83.5	17.40	87.0	87.6
14.42	75.1	75.6	15.42	79.1	79.6	16.42	83.0	83.6	17.42	87.1	87.7
14.44	75.2	75.7	15.44	79.1	79.7	16.44	83.1	83.7	17.44	87.2	87.8
14.46	75.3	75.8	15.46	79.2	79.7	16.46	83.2	83.8	17.46	87.3	87.9
14.48	75.4	75.9	15.48	79.3	79.8	16.48	83.3	83.8	17.48	87.3	87.9
14.50	75.4	76.0	15.50	79.4	79.9	16.50	83.4	83.9	17.50	87.4	88.0
14.52	75.5	76.0	15.52	79.5	80.0	16.52	83.5	84.0	17.52	87.5	88.1
14.54	75.6	76.1	15.54	79.5	80.1	16.54	83.5	84.1	17.54	87.6	88.2
14.56	75.7	76.2	15.56	79.6	80.1	16.56	83.6	84.2	17.56	87.7	88.3
14.58	75.8	76.3	15.58	79.7	80.2	16.58	83.7	84.3	17.58	87.8	88.3
14.60	75.8	76.3	15.60	79.8	80.3	16.60	83.8	84.3	17.60	87.8	88.4
14.62	75.9	76.4	15.62	79.8	80.4	16.62	83.9	84.4	17.62	87.9	88.5
14.64	76.0	76.5	15.64	79.9	80.5	16.64	83.9	84.5	17.64	88.0	88.6
14.66	76.1	76.6	15.66	80.0	80.5	16.66	84.0	84.6	17.66	88.1	88.7
14.68	76.1	76.7	15.68	80.1	80.6	16.68	84.1	84.7	17.68	88.2	88.8
14.70	76.2	76.7	15.70	80.2	80.7	16.70	84.2	84.7	17.70	88.3	88.8
14.72	76.3	76.8	15.72	80.2	80.8	16.72	84.3	84.8	17.72	88.3	88.9
14.74	76.4	76.9	15.74	80.3	80.9	16.74	84.3	84.9	17.74	88.4	89.0
14.76	76.5	77.0	15.76	80.4	80.9	16.76	84.4	85.0	17.76	88.5	89.1
14.78	76.5	77.1	15.78	80.5	81.0	16.78	84.5	85.1	17.78	88.6	89.2
14.80	76.6	77.1	15.80	80.6	81.1	16.80	84.6	85.1	17.80	88.7	89.3
14.82	76.7	77.2	15.82	80.6	81.2	16.82	84.7	85.2	17.82	88.7	89.3
14.84	76.8	77.3	15.84	80.7	81.3	16.84	84.7	85.3	17.84	88.8	89.4
14.86	76.9	77.4	15.86	80.8	81.3	16.86	84.8	85.4	17.86	88.9	89.5
14.88	76.9	77.4	15.88	80.9	81.4	16.88	84.9	85.5	17.88	89.0	89.6
14.90	77.0	77.5	15.90	81.0	81.5	16.90	85.0	85.6	17.90	89.1	89.7
14.92	77.1	77.6	15.92	81.0	81.6	16.92	85.1	85.6	17.92	89.2	89.8
14.94	77.2	77.7	15.94	81.1	81.7	16.94	85.1	85.7	17.94	89.2	89.8
14.96	77.2	77.8	15.96	81.2	81.7	16.96	85.2	85.8	17.96	89.3	89.9
14.98	77.3	77.8	15.98	81.3	81.8	16.98	85.3	85.9	17.98	89.4	90.0

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
18.00	89.5	90.1	19.00	93.6	94.3	20.00	97.8	98.5	22.50	108.5	109.2
18.02	89.6	90.2	19.02	93.7	94.3	20.05	98.0	98.7	22.55	108.7	109.4
18.04	89.6	90.2	19.04	93.8	94.4	20.10	98.2	98.9	22.60	108.9	109.6
18.06	89.7	90.3	19.06	93.9	94.5	20.15	98.5	99.1	22.65	109.1	109.9
18.08	89.8	90.4	19.08	94.0	94.6	20.20	98.7	99.3	22.70	109.4	110.1
18.10	89.9	90.5	19.10	94.0	94.7	20.25	98.9	99.5	22.75	109.6	110.3
18.12	90.0	90.6	19.12	94.1	94.8	20.30	99.1	99.8	22.80	109.8	110.5
18.14	90.1	90.7	19.14	94.2	94.8	20.35	99.3	100.0	22.85	110.0	110.7
18.16	90.1	90.7	19.16	94.3	94.9	20.40	99.5	100.2	22.90	110.2	111.0
18.18	90.2	90.8	19.18	94.4	95.0	20.45	99.7	100.4	22.95	110.4	111.2
18.20	90.3	90.9	19.20	94.5	95.1	20.50	99.9	100.6	23.00	110.6	111.4
18.22	90.4	91.0	19.22	94.5	95.2	20.55	100.1	100.8	23.05	110.9	111.6
18.24	90.5	91.1	19.24	94.6	95.3	20.60	100.4	101.0	23.10	111.1	111.8
18.26	90.6	91.2	19.26	94.7	95.4	20.65	100.6	101.2	23.15	111.3	112.0
18.28	90.6	91.2	19.28	94.8	95.4	20.70	100.8	101.5	23.20	111.5	112.3
18.30	90.7	91.3	19.30	94.9	95.5	20.75	101.0	101.7	23.25	111.7	112.5
18.32	90.8	91.4	19.32	95.0	95.6	20.80	101.2	101.9	23.30	111.9	112.7
18.34	90.9	91.5	19.34	95.0	95.7	20.85	101.4	102.1	23.35	112.2	112.9
18.36	91.0	91.6	19.36	95.1	95.8	20.90	101.6	102.3	23.40	112.4	113.1
18.38	91.1	91.7	19.38	95.2	95.9	20.95	101.8	102.5	23.45	112.6	113.4
18.40	91.1	91.7	19.40	95.3	95.9	21.00	102.1	102.7	23.50	112.8	113.6
18.42	91.2	91.8	19.42	95.4	96.0	21.05	102.3	103.0	23.55	113.0	113.8
18.44	91.3	91.9	19.44	95.5	96.1	21.10	102.5	103.2	23.60	113.2	114.0
18.46	91.4	92.0	19.46	95.6	96.2	21.15	102.7	103.4	23.65	113.5	114.2
18.48	91.5	92.1	19.48	95.6	96.3	21.20	102.9	103.6	23.70	113.7	114.4
18.50	91.5	92.2	19.50	95.7	96.4	21.25	103.1	103.8	23.75	113.9	114.7
18.52	91.6	92.2	19.52	95.8	96.4	21.30	103.3	104.0	23.80	114.1	114.9
18.54	91.7	92.3	19.54	95.9	96.5	21.35	103.6	104.2	23.85	114.3	115.1
18.56	91.8	92.4	19.56	96.0	96.6	21.40	103.8	104.5	23.90	114.6	115.3
18.58	91.9	92.5	19.58	96.1	96.7	21.45	104.0	104.7	23.95	114.8	115.5
18.60	92.0	92.6	19.60	96.1	96.8	21.50	104.2	104.9	24.00	115.0	115.8
18.62	92.0	92.7	19.62	96.2	96.9	21.55	104.4	105.1	24.05	115.2	116.0
18.64	92.1	92.7	19.64	96.3	97.0	21.60	104.6	105.3	24.10	115.4	116.2
18.66	92.2	92.8	19.66	96.4	97.0	21.65	104.8	105.5	24.15	115.6	116.4
18.68	92.3	92.9	19.68	96.5	97.1	21.70	105.0	105.8	24.20	115.9	116.6
18.70	92.4	93.0	19.70	96.6	97.2	21.75	105.3	106.0	24.25	116.1	116.9
18.72	92.5	93.1	19.72	96.6	97.3	21.80	105.5	106.2	24.30	116.3	117.1
18.74	92.5	93.2	19.74	96.7	97.4	21.85	105.7	106.4	24.35	116.4	117.3
18.76	92.6	93.3	19.76	96.8	97.5	21.90	105.9	106.6	24.40	116.7	117.5
18.78	92.7	93.3	19.78	96.9	97.5	21.95	106.1	106.8	24.45	117.0	117.7
18.80	92.8	93.4	19.80	97.0	97.6	22.00	106.3	107.0	24.50	117.2	118.0
18.82	92.9	93.5	19.82	97.1	97.7	22.05	106.6	107.3	24.55	117.4	118.2
18.84	93.0	93.6	19.84	97.1	97.8	22.10	106.8	107.5	24.60	117.6	118.4
18.86	93.0	93.7	19.86	97.2	97.9	22.15	107.0	107.7	24.65	117.8	118.6
18.88	93.1	93.8	19.88	97.3	98.0	22.20	107.2	107.9	24.70	118.0	118.8
18.90	93.2	93.8	19.90	97.4	98.1	22.25	107.4	108.1	24.75	118.3	119.1
18.92	93.3	93.9	19.92	97.5	98.1	22.30	107.6	108.3	24.80	118.5	119.3
18.94	93.4	94.0	19.94	97.6	98.2	22.35	107.8	108.6	24.85	118.7	119.5
18.96	93.5	94.1	19.96	97.7	98.3	22.40	108.1	108.8	24.90	118.9	119.7
18.98	93.5	94.2	19.98	97.7	98.4	22.45	108.3	109.0	24.95	119.1	119.9

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
25.00	119.4	120.2	27.50	130.4	131.3	30.00	141.5	142.5	32.50	152.7	153.8
25.05	119.6	120.4	27.55	130.6	131.5	30.05	141.7	142.7	32.55	153.0	154.0
25.10	119.8	120.6	27.60	130.8	131.7	30.10	142.0	142.9	32.60	153.2	154.2
25.15	120.0	120.8	27.65	131.0	131.9	30.15	142.2	143.1	32.65	153.4	154.4
25.20	120.2	121.0	27.70	131.3	132.1	30.20	142.4	143.4	32.70	153.6	154.7
25.25	120.5	121.3	27.75	131.5	132.4	30.25	142.6	143.6	32.75	153.9	154.9
25.30	120.7	121.5	27.80	131.7	132.6	30.30	142.9	143.8	32.80	154.1	155.1
25.35	120.9	121.7	27.85	131.9	132.8	30.35	143.1	144.0	32.85	154.3	155.4
25.40	121.1	121.9	27.90	132.2	133.0	30.40	143.3	144.3	32.90	154.5	155.6
25.45	121.3	122.1	27.95	132.4	133.3	30.45	143.5	144.5	32.95	154.8	155.8
25.50	121.6	122.4	28.00	132.6	133.5	30.50	143.8	144.7	33.00	155.0	156.0
25.55	121.8	122.6	28.05	132.8	133.7	30.55	144.0	144.9	33.05	155.2	156.3
25.60	122.0	122.8	28.10	133.0	133.9	30.60	144.2	145.2	33.10	155.4	156.5
25.65	122.2	123.0	28.15	133.3	134.2	30.65	144.4	145.4	33.15	155.7	156.7
25.70	122.4	123.3	28.20	133.5	134.4	30.70	144.6	145.6	33.20	155.9	156.9
25.75	122.6	123.5	28.25	133.7	134.6	30.75	144.9	145.8	33.25	156.1	157.2
25.80	122.9	123.7	28.30	133.9	134.8	30.80	145.1	146.1	33.30	156.3	157.4
25.85	123.1	123.9	28.35	134.2	135.1	30.85	145.3	146.3	33.35	156.6	157.6
25.90	123.3	124.1	28.40	134.4	135.3	30.90	145.5	146.5	33.40	156.8	157.8
25.95	123.5	124.4	28.45	134.6	135.5	30.95	145.8	146.7	33.45	157.0	158.1
26.00	123.7	124.6	28.50	134.8	135.7	31.00	146.0	147.0	33.50	157.2	158.3
26.05	124.0	124.8	28.55	135.0	135.9	31.05	146.2	147.2	33.55	157.5	158.5
26.10	124.2	125.0	28.60	135.3	136.2	31.10	146.4	147.4	33.60	157.7	158.8
26.15	124.4	125.2	28.65	135.5	136.4	31.15	146.7	147.7	33.65	157.9	159.0
26.20	124.6	125.5	28.70	135.7	136.6	31.20	146.9	147.9	33.70	158.2	159.2
26.25	124.9	125.7	28.75	135.9	136.8	31.25	147.1	148.1	33.75	158.4	159.4
26.30	125.1	125.9	28.80	136.2	137.1	31.30	147.3	148.3	33.80	158.6	159.7
26.35	125.3	126.1	28.85	136.4	137.3	31.35	147.6	148.6	33.85	158.8	159.9
26.40	125.5	126.4	28.90	136.6	137.5	31.40	147.8	148.8	33.90	159.1	160.1
26.45	125.7	126.6	28.95	136.8	137.7	31.45	148.0	149.0	33.95	159.3	160.3
26.50	126.0	126.8	29.00	137.0	138.0	31.50	148.2	149.2	34.00	159.5	160.6
26.55	126.2	127.0	29.05	137.3	138.2	31.55	148.5	149.5	34.05	159.7	160.8
26.60	126.4	127.2	29.10	137.5	138.4	31.60	148.7	149.7	34.10	160.0	161.0
26.65	126.6	127.5	29.15	137.7	138.6	31.65	148.9	149.9	34.15	160.2	161.3
26.70	126.8	127.7	29.20	137.9	138.9	31.70	149.1	150.1	34.20	160.4	161.5
26.75	127.1	127.9	29.25	138.2	139.1	31.75	149.4	150.4	34.25	160.6	161.7
26.80	127.3	128.1	29.30	138.4	139.3	31.80	149.6	150.6	34.30	160.9	161.9
26.85	127.5	128.4	29.35	138.6	139.5	31.85	149.8	150.8	34.35	161.1	162.2
26.90	127.7	128.6	29.40	138.8	139.8	31.90	150.0	151.0	34.40	161.3	162.4
26.95	127.9	128.8	29.45	139.1	140.0	31.95	150.3	151.3	34.45	161.5	162.6
27.00	128.2	129.0	29.50	139.3	140.2	32.00	150.5	151.5	34.50	161.8	162.9
27.05	128.4	129.2	29.55	139.5	140.4	32.05	150.7	151.7	34.55	162.0	163.1
27.10	128.6	129.5	29.60	139.7	140.7	32.10	150.9	152.0	34.60	162.2	163.3
27.15	128.8	129.7	29.65	140.0	140.9	32.15	151.2	152.2	34.65	162.4	163.5
27.20	129.0	129.9	29.70	140.2	141.1	32.20	151.4	152.4	34.70	162.7	163.8
27.25	129.3	130.1	29.75	140.4	141.3	32.25	151.6	152.6	34.75	162.9	164.0
27.30	129.5	130.4	29.80	140.6	141.6	32.30	151.8	152.9	34.80	163.1	164.2
27.35	129.7	130.6	29.85	140.8	141.8	32.35	152.1	153.1	34.85	163.3	164.4
27.40	129.9	130.8	29.90	141.1	142.0	32.40	152.3	153.3	34.90	163.6	164.7
27.45	130.2	131.0	29.95	141.3	142.2	32.45	152.5	153.5	34.95	163.8	164.9

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
35.00	164.0	165.1	37.50	175.4	176.5	40.00	186.8	188.0	42.50	198.2	199.5
35.05	164.3	165.4	37.55	175.6	176.8	40.05	187.0	188.2	42.55	198.4	199.7
35.10	164.5	165.6	37.60	175.8	177.0	40.10	187.2	188.5	42.60	198.6	200
35.15	164.7	165.8	37.65	176.1	177.2	40.15	187.4	188.7	42.65	198.9	200
35.20	164.9	166.0	37.70	176.3	177.5	40.20	187.7	188.9	42.70	199.1	200
35.25	165.2	166.3	37.75	176.5	177.7	40.25	187.9	189.2	42.75	199.3	201
35.30	165.4	166.5	37.80	176.7	177.9	40.30	188.1	189.4	42.80	199.5	201
35.35	165.6	166.7	37.85	177.0	178.1	40.35	188.4	189.6	42.85	199.8	201
35.40	165.8	167.0	37.90	177.2	178.4	40.40	188.6	189.8	42.90	200	201
35.45	166.1	167.2	37.95	177.4	178.6	40.45	188.8	190.1	42.95	200	202
35.50	166.3	167.4	38.00	177.6	178.8	40.50	189.0	190.3	43.00	200	202
35.55	166.5	167.6	38.05	177.9	179.1	40.55	189.3	190.5	43.05	201	202
35.60	166.7	167.9	38.10	178.1	179.3	40.60	189.5	190.8	43.10	201	202
35.65	167.0	168.1	38.15	178.3	179.5	40.65	189.7	191.0	43.15	201	202
35.70	167.2	168.3	38.20	178.6	179.8	40.70	189.9	191.2	43.20	201	203
35.75	167.4	168.6	38.25	178.8	180.0	40.75	190.2	191.5	43.25	202	203
35.80	167.7	168.8	38.30	179.0	180.2	40.80	190.4	191.7	43.30	202	203
35.85	167.9	169.0	38.35	179.2	180.4	40.85	190.6	191.9	43.35	202	203
35.90	168.1	169.2	38.40	179.5	180.7	40.90	190.9	192.1	43.40	202	203
35.95	168.3	169.5	38.45	179.7	180.9	40.95	191.1	192.4	43.45	203	204
36.00	168.6	169.7	38.50	179.9	181.1	41.00	191.3	192.6	43.50	203	204
36.05	168.8	169.9	38.55	180.1	181.4	41.05	191.5	192.8	43.55	203	204
36.10	169.0	170.1	38.60	180.4	181.6	41.10	191.8	193.1	43.60	203	204
36.15	169.2	170.4	38.65	180.6	181.8	41.15	192.0	193.3	43.65	203	205
36.20	169.5	170.6	38.70	180.8	182.0	41.20	192.2	193.5	43.70	204	205
36.25	169.7	170.8	38.75	181.1	182.3	41.25	192.5	193.7	43.75	204	205
36.30	169.9	171.1	38.80	181.3	182.5	41.30	192.7	194.0	43.80	204	205
36.35	170.1	171.3	38.85	181.5	182.7	41.35	192.9	194.2	43.85	204	206
36.40	170.4	171.5	38.90	181.7	183.0	41.40	193.1	194.4	43.90	205	206
36.45	170.6	171.7	38.95	182.0	183.2	41.45	193.4	194.7	43.95	205	206
36.50	170.8	172.0	39.00	182.2	183.4	41.50	193.6	194.9	44.00	205	206
36.55	171.1	172.2	39.05	182.4	183.6	41.55	193.8	195.1	44.05	205	207
36.60	171.3	172.4	39.10	182.7	183.9	41.60	194.1	195.4	44.10	205	207
36.65	171.5	172.7	39.15	182.9	184.1	41.65	194.3	195.6	44.15	206	207
36.70	171.7	172.9	39.20	183.1	184.3	41.70	194.5	195.8	44.20	206	207
36.75	172.0	173.1	39.25	183.3	184.6	41.75	194.7	196.0	44.25	206	208
36.80	172.2	173.3	39.30	183.6	184.8	41.80	195.0	196.3	44.30	206	208
36.85	172.4	173.6	39.35	183.8	185.0	41.85	195.2	196.5	44.35	207	208
36.90	172.6	173.8	39.40	184.0	185.3	41.90	195.4	196.7	44.40	207	208
36.95	172.9	174.0	39.45	184.2	185.5	41.95	195.7	197.0	44.45	207	208
37.00	173.1	174.3	39.50	184.5	185.7	42.00	195.9	197.2	44.50	207	209
37.05	173.3	174.5	39.55	184.7	185.9	42.05	196.1	197.4	44.55	208	209
37.10	173.6	174.7	39.60	184.9	186.2	42.10	196.3	197.7	44.60	208	209
37.15	173.8	174.9	39.65	185.2	186.4	42.15	196.6	197.9	44.65	208	209
37.20	174.0	175.2	39.70	185.4	186.6	42.20	196.8	198.1	44.70	208	210
37.25	174.2	175.4	39.75	185.6	186.9	42.25	197.0	198.3	44.75	208	210
37.30	174.5	175.6	39.80	185.8	187.1	42.30	197.3	198.6	44.80	209	210
37.35	174.7	175.9	39.85	186.1	187.3	42.35	197.5	198.8	44.85	209	210
37.40	174.9	176.1	39.90	186.3	187.5	42.40	197.7	199.0	44.90	209	211
37.45	175.1	176.3	39.95	186.5	187.8	42.45	197.9	199.3	44.95	209	211

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
45.00	210	211	47.50	221	223	50.0	233	234	55.0	256	257
45.05	210	211	47.55	221	223	50.1	233	235	55.1	256	258
45.10	210	211	47.60	222	223	50.2	233	235	55.2	256	258
45.15	210	212	47.65	222	223	50.3	234	235	55.3	257	259
45.20	211	212	47.70	222	223	50.4	234	236	55.4	257	259
45.25	211	212	47.75	222	224	50.5	235	236	55.5	258	260
45.30	211	212	47.80	222	224	50.6	235	237	55.6	258	260
45.35	211	213	47.85	223	224	50.7	236	237	55.7	259	261
45.40	211	213	47.90	223	224	50.8	236	238	55.8	259	261
45.45	212	213	47.95	223	225	50.9	237	238	55.9	260	261
45.50	212	213	48.00	223	225	51.0	237	239	56.0	260	262
45.55	212	214	48.05	224	225	51.1	238	239	56.1	261	262
45.60	212	214	48.10	224	225	51.2	238	240	56.2	261	263
45.65	213	214	48.15	224	226	51.3	239	240	56.3	262	263
45.70	213	214	48.20	224	226	51.4	239	241	56.4	262	264
45.75	213	214	48.25	225	226	51.5	239	241	56.5	262	264
45.80	213	215	48.30	225	226	51.6	240	242	56.6	263	265
45.85	214	215	48.35	225	226	51.7	240	242	56.7	263	265
45.90	214	215	48.40	225	227	51.8	241	242	56.8	264	266
45.95	214	215	48.45	225	227	51.9	241	243	56.9	264	266
46.00	214	216	48.50	226	227	52.0	242	243	57.0	265	267
46.05	214	216	48.55	226	227	52.1	242	244	57.1	265	267
46.10	215	216	48.60	226	228	52.2	243	244	57.2	266	267
46.15	215	216	48.65	226	228	52.3	243	245	57.3	266	268
46.20	215	217	48.70	227	228	52.4	244	245	57.4	267	268
46.25	215	217	48.75	227	228	52.5	244	246	57.5	267	269
46.30	216	217	48.80	227	229	52.6	245	246	57.6	268	269
46.35	216	217	48.85	227	229	52.7	245	247	57.7	268	270
46.40	216	217	48.90	227	229	52.8	245	247	57.8	268	270
46.45	216	218	48.95	228	229	52.9	246	248	57.9	269	271
46.50	216	218	49.00	228	229	53.0	246	248	58.0	269	271
46.55	217	218	49.05	228	230	53.1	247	248	58.1	270	272
46.60	217	218	49.10	228	230	53.2	247	249	58.2	270	272
46.65	217	219	49.15	229	230	53.3	248	249	58.3	271	273
46.70	217	219	49.20	229	230	53.4	248	250	58.4	271	273
46.75	218	219	49.25	229	231	53.5	249	250	58.5	272	273
46.80	218	219	49.30	229	231	53.6	249	251	58.6	272	274
46.85	218	220	49.35	230	231	53.7	250	251	58.7	273	274
46.90	218	220	49.40	230	231	53.8	250	252	58.8	273	275
46.95	219	220	49.45	230	232	53.9	250	252	58.9	274	275
47.00	219	220	49.50	230	232	54.0	251	253	59.0	274	276
47.05	219	220	49.55	230	232	54.1	251	253	59.1	274	276
47.10	219	221	49.60	231	232	54.2	252	254	59.2	275	277
47.15	219	221	49.65	231	232	54.3	252	254	59.3	275	277
47.20	220	221	49.70	231	233	54.4	253	254	59.4	276	278
47.25	220	221	49.75	231	233	54.5	253	255	59.5	276	278
47.30	220	222	49.80	232	233	54.6	254	255	59.6	277	279
47.35	220	222	49.85	232	233	54.7	254	256	59.7	277	279
47.40	221	222	49.90	232	234	54.8	255	256	59.8	278	280
47.45	221	222	49.95	232	234	54.9	255	257	59.9	278	280

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
60.0	279	280	65.0	302	304	70.0	325	327	75.0	348	350
60.1	279	281	65.1	302	304	70.1	325	327	75.1	348	351
60.2	280	281	65.2	303	305	70.2	326	328	75.2	349	351
60.3	280	282	65.3	303	305	70.3	326	328	75.3	349	352
60.4	280	282	65.4	303	306	70.4	327	329	75.4	350	352
60.5	281	283	65.5	304	306	70.5	327	329	75.5	350	352
60.6	281	283	65.6	304	306	70.6	328	330	75.6	351	353
60.7	282	284	65.7	305	307	70.7	328	330	75.7	351	353
60.8	282	284	65.8	305	307	70.8	328	331	75.8	352	354
60.9	283	285	65.9	306	308	70.9	329	331	75.9	352	354
61.0	283	285	66.0	306	308	71.0	329	332	76.0	352	355
61.1	284	286	66.1	307	309	71.1	330	332	76.1	353	355
61.2	284	286	66.2	307	309	71.2	330	332	76.2	353	356
61.3	285	286	66.3	308	310	71.3	331	333	76.3	354	356
61.4	285	287	66.4	308	310	71.4	331	333	76.4	354	357
61.5	286	287	66.5	309	311	71.5	332	334	76.5	355	357
61.6	286	288	66.6	309	311	71.6	332	334	76.6	355	358
61.7	286	288	66.7	309	312	71.7	333	335	76.7	356	358
61.8	287	289	66.8	310	312	71.8	333	335	76.8	356	359
61.9	287	289	66.9	310	313	71.9	334	336	76.9	357	359
62.0	288	290	67.0	311	313	72.0	334	336	77.0	357	359
62.1	288	290	67.1	311	313	72.1	334	337	77.1	358	360
62.2	289	291	67.2	312	314	72.2	335	337	77.2	358	360
62.3	289	291	67.3	312	314	72.3	335	338	77.3	358	361
62.4	290	292	67.4	313	315	72.4	336	338	77.4	359	361
62.5	290	292	67.5	313	315	72.5	336	339	77.5	359	362
62.6	291	293	67.6	314	316	72.6	337	339	77.6	360	362
62.7	291	293	67.7	314	316	72.7	337	339	77.7	360	363
62.8	291	293	67.8	315	317	72.8	338	340	77.8	361	363
62.9	292	294	67.9	315	317	72.9	338	340	77.9	361	364
63.0	292	294	68.0	315	318	73.0	339	341	78.0	362	364
63.1	293	295	68.1	316	318	73.1	339	341	78.1	362	365
63.2	293	295	68.2	316	319	73.2	340	342	78.2	363	365
63.3	294	296	68.3	317	319	73.3	340	342	78.3	363	366
63.4	294	296	68.4	317	319	73.4	340	343	78.4	364	366
63.5	295	297	68.5	318	320	73.5	341	343	78.5	364	366
63.6	295	297	68.6	318	320	73.6	341	344	78.6	364	367
63.7	296	298	68.7	319	321	73.7	342	344	78.7	365	367
63.8	296	298	68.8	319	321	73.8	342	345	78.8	365	368
63.9	297	299	68.9	320	322	73.9	343	345	78.9	366	368
64.0	297	299	69.0	320	322	74.0	343	346	79.0	366	369
64.1	297	299	69.1	321	323	74.1	344	346	79.1	367	369
64.2	298	300	69.2	321	323	74.2	344	346	79.2	367	370
64.3	298	300	69.3	322	324	74.3	345	347	79.3	368	370
64.4	299	301	69.4	322	324	74.4	345	347	79.4	368	371
64.5	299	301	69.5	322	325	74.5	346	348	79.5	369	371
64.6	300	302	69.6	323	325	74.6	346	348	79.6	369	372
64.7	300	302	69.7	323	326	74.7	346	349	79.7	370	372
64.8	301	303	69.8	324	326	74.8	347	349	79.8	370	373
64.9	301	303	69.9	324	326	74.9	347	350	79.9	370	373

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
80.0	371	373	85.0	394	397	90.0	417	420	95.0	440	443
80.1	371	374	85.1	395	397	90.1	418	420	95.1	441	444
80.2	372	374	85.2	395	398	90.2	418	421	95.2	441	444
80.3	372	375	85.3	395	398	90.3	419	421	95.3	442	445
80.4	373	375	85.4	396	399	90.4	419	422	95.4	442	445
80.5	373	376	85.5	396	399	90.5	420	422	95.5	443	446
80.6	374	376	85.6	397	400	90.6	420	423	95.6	443	446
80.7	374	377	85.7	397	400	90.7	420	423	95.7	444	447
80.8	375	377	85.8	398	400	90.8	421	424	95.8	444	447
80.9	375	378	85.9	398	401	90.9	421	424	95.9	444	447
81.0	376	378	86.0	399	401	91.0	422	425	96.0	445	448
81.1	376	379	86.1	399	402	91.1	422	425	96.1	445	448
81.2	376	379	86.2	400	402	91.2	423	426	96.2	446	449
81.3	377	379	86.3	400	403	91.3	423	426	96.3	446	449
81.4	377	380	86.4	401	403	91.4	424	427	96.4	447	450
81.5	378	380	86.5	401	404	91.5	424	427	96.5	447	450
81.6	378	381	86.6	401	404	91.6	425	427	96.6	448	451
81.7	379	381	86.7	402	405	91.7	425	428	96.7	448	451
81.8	379	382	86.8	402	405	91.8	426	428	96.8	449	452
81.9	380	382	86.9	403	406	91.9	426	429	96.9	449	452
82.0	380	383	87.0	403	406	92.0	426	429	97.0	450	453
82.1	381	383	87.1	404	406	92.1	427	430	97.1	450	453
82.2	381	383	87.2	404	407	92.2	427	430	97.2	451	454
82.3	382	384	87.3	405	407	92.3	428	431	97.3	451	454
82.4	382	385	87.4	405	408	92.4	428	431	97.4	451	454
82.5	383	385	87.5	406	408	92.5	429	432	97.5	452	455
82.6	383	386	87.6	406	409	92.6	429	432	97.6	452	455
82.7	383	386	87.7	407	409	92.7	430	433	97.7	453	456
82.8	384	386	87.8	407	410	92.8	430	433	97.8	453	456
82.9	384	387	87.9	407	410	92.9	431	433	97.9	454	457
83.0	385	387	88.0	408	411	93.0	431	434	98.0	454	457
83.1	385	388	88.1	408	411	93.1	432	434	98.1	455	458
83.2	386	388	88.2	409	412	93.2	432	435	98.2	455	458
83.3	386	389	88.3	409	412	93.3	432	435	98.3	456	459
83.4	387	389	88.4	410	413	93.4	433	436	98.4	456	459
83.5	387	390	88.5	410	413	93.5	433	436	98.5	457	460
83.6	388	390	88.6	411	413	93.6	434	437	98.6	457	460
83.7	388	391	88.7	411	414	93.7	434	437	98.7	457	461
83.8	389	391	88.8	412	414	93.8	435	438	98.8	458	461
83.9	389	392	88.9	412	415	93.9	435	438	98.9	458	461
84.0	389	392	89.0	413	415	94.0	436	439	99.0	459	462
84.1	390	393	89.1	413	416	94.1	436	439	99.1	459	462
84.2	390	392	89.2	413	416	94.2	437	440	99.2	460	463
84.3	391	393	89.3	414	417	94.3	437	440	99.3	460	463
84.4	391	394	89.4	414	417	94.4	438	440	99.4	461	464
84.5	392	394	89.5	415	418	94.5	438	441	99.5	461	464
84.6	392	395	89.6	415	418	94.6	438	441	99.6	462	465
84.7	393	395	89.7	416	419	94.7	439	442	99.7	462	465
84.8	393	396	89.8	416	419	94.8	439	442	99.8	463	466
84.9	394	396	89.9	417	420	94.9	440	443	99.9	463	466

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS		Kin Vis, mm ² /s (cSt)	Equivalent Saybolt Universal Viscosity, SUS	
	At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F		At 100°F	At 210°F
100.0	463	467	110.0	510	513	120.0	556	560	130.0	602	606
100.2	464	468	110.2	511	514	120.2	557	561	130.2	603	607
100.4	465	468	110.4	512	515	120.4	558	562	130.4	604	608
100.6	466	469	110.6	513	516	120.6	559	563	130.6	605	609
100.8	467	470	110.8	513	517	120.8	560	564	130.8	606	610
101.0	468	471	111.0	514	518	121.0	561	564	131.0	607	611
101.2	469	472	111.2	515	519	121.2	562	565	131.2	608	612
101.4	470	473	111.4	516	520	121.4	563	566	131.4	609	613
101.6	471	474	111.6	517	521	121.6	563	567	131.6	610	614
101.8	472	475	111.8	518	522	121.8	564	568	131.8	611	615
102.0	473	476	112.0	519	522	122.0	565	569	132.0	612	616
102.2	474	477	112.2	520	523	122.2	566	570	132.2	613	617
102.4	475	478	112.4	521	524	122.4	567	571	132.4	613	618
102.6	475	479	112.6	522	525	122.6	568	572	132.6	614	619
102.8	476	480	112.8	523	526	122.8	569	573	132.8	615	619
103.0	477	481	113.0	524	527	123.0	570	574	133.0	616	620
103.2	478	481	113.2	525	528	123.2	571	575	133.2	617	621
103.4	479	482	113.4	525	529	123.4	572	576	133.4	618	622
103.6	480	483	113.6	526	530	123.6	573	577	133.6	619	623
103.8	481	484	113.8	527	531	123.8	574	577	133.8	620	624
104.0	482	485	114.0	528	532	124.0	575	578	134.0	621	625
104.2	483	486	114.2	529	533	124.2	575	579	134.2	622	626
104.4	484	487	114.4	530	534	124.4	576	580	134.4	623	627
104.6	485	488	114.6	531	535	124.6	577	581	134.6	624	628
104.8	486	489	114.8	532	536	124.8	578	582	134.8	625	629
105.0	487	490	115.0	533	536	125.0	579	583	135.0	625	630
105.2	488	491	115.2	534	537	125.2	580	584	135.2	626	631
105.4	488	492	115.4	535	538	125.4	581	585	135.4	627	632
105.6	489	493	115.6	536	539	125.6	582	586	135.6	628	632
105.8	490	494	115.8	537	540	125.8	583	587	135.8	629	633
106.0	491	495	116.0	538	541	126.0	584	588	136.0	630	634
106.2	492	495	116.2	538	542	126.2	585	589	136.2	631	635
106.4	493	496	116.4	539	543	126.4	586	590	136.4	632	636
106.6	494	497	116.6	540	544	126.6	587	591	136.6	633	637
106.8	495	498	116.8	541	545	126.8	588	591	136.8	634	638
107.0	496	499	117.0	542	546	127.0	588	592	137.0	635	639
107.2	497	500	117.2	543	547	127.2	589	593	137.2	636	640
107.4	498	501	117.4	544	548	127.4	590	594	137.4	637	641
107.6	499	502	117.6	545	549	127.6	591	595	137.6	638	642
107.8	500	503	117.8	546	550	127.8	592	596	137.8	638	643
108.0	500	504	118.0	547	550	128.0	593	597	138.0	639	644
108.2	501	505	118.2	548	551	128.2	594	598	138.2	640	645
108.4	502	506	118.4	549	552	128.4	595	599	138.4	641	646
108.6	503	507	118.6	550	553	128.6	596	600	138.6	642	646
108.8	504	508	118.8	550	554	128.8	597	601	138.8	643	647
109.0	505	509	119.0	551	555	129.0	598	602	139.0	644	648
109.2	506	509	119.2	552	556	129.2	599	603	139.2	645	649
109.4	507	510	119.4	553	557	129.4	600	604	139.4	646	650
109.6	508	511	119.6	554	558	129.6	600	605	139.6	647	651
109.8	509	512	119.8	555	559	129.8	601	605	139.8	648	652

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F
140.0	649	150.0	695	160.0	741	170.0	788	180.0	834	190.0	880
140.2	650	150.2	696	160.2	742	170.2	789	180.2	835	190.2	881
140.4	650	150.4	697	160.4	743	170.4	789	180.4	836	190.4	882
140.6	651	150.6	698	160.6	744	170.6	790	180.6	837	190.6	883
140.8	652	150.8	699	160.8	745	170.8	791	180.8	838	190.8	884
141.0	653	151.0	700	161.0	746	171.0	792	181.0	839	191.0	885
141.2	654	151.2	701	161.2	747	171.2	793	181.2	839	191.2	886
141.4	655	151.4	701	161.4	748	171.4	794	181.4	840	191.4	887
141.6	656	151.6	702	161.6	749	171.6	795	181.6	841	191.6	888
141.8	657	151.8	703	161.8	750	171.8	796	181.8	842	191.8	889
142.0	658	152.0	704	162.0	751	172.0	797	182.0	843	192.0	889
142.2	659	152.2	705	162.2	751	172.2	798	182.2	844	192.2	890
142.4	660	152.4	706	162.4	752	172.4	799	182.4	845	192.4	891
142.6	661	152.6	707	162.6	753	172.6	800	182.6	846	192.6	892
142.8	662	152.8	708	162.8	754	172.8	801	182.8	847	192.8	893
143.0	663	153.0	709	163.0	755	173.0	801	183.0	848	193.0	894
143.2	663	153.2	710	163.2	756	173.2	802	183.2	849	193.2	895
143.4	664	153.4	711	163.4	757	173.4	803	183.4	850	193.4	896
143.6	665	153.6	712	163.6	758	173.6	804	183.6	851	193.6	897
143.8	666	153.8	713	163.8	759	173.8	805	183.8	851	193.8	898
144.0	667	154.0	713	164.0	760	174.0	806	184.0	852	194.0	899
144.2	668	154.2	714	164.2	761	174.2	807	184.2	853	194.2	900
144.4	669	154.4	715	164.4	762	174.4	808	184.4	854	194.4	901
144.6	670	154.6	716	164.6	763	174.6	809	184.6	855	194.6	902
144.8	671	154.8	717	164.8	763	174.8	810	184.8	856	194.8	902
145.0	672	155.0	718	165.0	764	175.0	811	185.0	857	195.0	903
145.2	673	155.2	719	165.2	765	175.2	812	185.2	858	195.2	904
145.4	674	155.4	720	165.4	766	175.4	813	185.4	859	195.4	905
145.6	675	155.6	721	165.6	767	175.6	814	185.6	860	195.6	906
145.8	676	155.8	722	165.8	768	175.8	814	185.8	861	195.8	907
146.0	676	156.0	723	166.0	769	176.0	815	186.0	862	196.0	908
146.2	677	156.2	724	166.2	770	176.2	816	186.2	863	196.2	909
146.4	678	156.4	725	166.4	771	176.4	817	186.4	864	196.4	910
146.6	679	156.6	726	166.6	772	176.6	818	186.6	864	196.6	911
146.8	680	156.8	726	166.8	773	176.8	819	186.8	865	196.8	912
147.0	681	157.0	727	167.0	774	177.0	820	187.0	866	197.0	913
147.2	682	157.2	728	167.2	775	177.2	821	187.2	867	197.2	914
147.4	683	157.4	729	167.4	776	177.4	822	187.4	868	197.4	914
147.6	684	157.6	730	167.6	776	177.6	823	187.6	869	197.6	915
147.8	685	157.8	731	167.8	777	177.8	824	187.8	870	197.8	916
148.0	686	158.0	732	168.0	778	178.0	825	188.0	871	198.0	917
148.2	687	158.2	733	168.2	779	178.2	826	188.2	872	198.2	918
148.4	688	158.4	734	168.4	780	178.4	826	188.4	873	198.4	919
148.6	688	158.6	735	168.6	781	178.6	827	188.6	874	198.6	920
148.8	689	158.8	736	168.8	782	178.8	828	188.8	875	198.8	921
149.0	690	159.0	737	169.0	783	179.0	829	189.0	876	199.0	922
149.2	691	159.2	738	169.2	784	179.2	830	189.2	877	199.2	923
149.4	692	159.4	738	169.4	785	179.4	831	189.4	877	199.4	924
149.6	693	159.6	739	169.6	786	179.6	832	189.6	878	199.6	925
149.8	694	159.8	740	169.8	787	179.8	833	189.8	879	199.8	926

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F
200.0	927	225.0	1042	250.0	1158	275.0	1274	300.0	1390	325.0	1506
200.5	929	225.5	1045	250.5	1160	275.5	1276	300.5	1392	325.5	1508
201.0	931	226.0	1047	251.0	1163	276.0	1279	301.0	1394	326.0	1510
201.5	933	226.5	1049	251.5	1165	276.5	1281	301.5	1397	326.5	1512
202.0	936	227.0	1052	252.0	1167	277.0	1283	302.0	1399	327.0	1515
202.5	938	227.5	1054	252.5	1170	277.5	1286	302.5	1401	327.5	1517
203.0	940	228.0	1056	253.0	1172	278.0	1288	303.0	1404	328.0	1519
203.5	943	228.5	1059	253.5	1174	278.5	1290	303.5	1406	328.5	1522
204.0	945	229.0	1061	254.0	1177	279.0	1292	304.0	1408	329.0	1524
204.5	947	229.5	1063	254.5	1179	279.5	1295	304.5	1411	329.5	1526
205.0	950	230.0	1065	255.0	1181	280.0	1297	305.0	1413	330.0	1529
205.5	952	230.5	1068	255.5	1184	280.5	1299	305.5	1415	330.5	1531
206.0	954	231.0	1070	256.0	1186	281.0	1302	306.0	1418	331.0	1533
206.5	957	231.5	1072	256.5	1188	281.5	1304	306.5	1420	331.5	1536
207.0	959	232.0	1075	257.0	1191	282.0	1306	307.0	1422	332.0	1538
207.5	961	232.5	1077	257.5	1193	282.5	1309	307.5	1424	332.5	1540
208.0	964	233.0	1079	258.0	1195	283.0	1311	308.0	1427	333.0	1543
208.5	966	233.5	1082	258.5	1198	283.5	1313	308.5	1429	333.5	1545
209.0	968	234.0	1084	259.0	1200	284.0	1316	309.0	1431	334.0	1547
209.5	971	234.5	1086	259.5	1202	284.5	1318	309.5	1434	334.5	1550
210.0	973	235.0	1089	260.0	1204	285.0	1320	310.0	1436	335.0	1552
210.5	975	235.5	1091	260.5	1207	285.5	1323	310.5	1438	335.5	1554
211.0	977	236.0	1093	261.0	1209	286.0	1325	311.0	1441	336.0	1557
211.5	980	236.5	1096	261.5	1211	286.5	1327	311.5	1443	336.5	1559
212.0	982	237.0	1098	262.0	1214	287.0	1330	312.0	1445	337.0	1561
212.5	984	237.5	1100	262.5	1216	287.5	1332	312.5	1448	337.5	1563
213.0	987	238.0	1103	263.0	1218	288.0	1334	313.0	1450	338.0	1566
213.5	989	238.5	1105	263.5	1221	288.5	1336	313.5	1452	338.5	1568
214.0	991	239.0	1107	264.0	1223	289.0	1339	314.0	1455	339.0	1570
214.5	994	239.5	1109	264.5	1225	289.5	1341	314.5	1457	339.5	1573
215.0	996	240.0	1112	265.0	1228	290.0	1343	315.0	1459	340.0	1575
215.5	998	240.5	1114	265.5	1230	290.5	1346	315.5	1462	340.5	1577
216.0	1001	241.0	1116	266.0	1232	291.0	1348	316.0	1464	341.0	1580
216.5	1003	241.5	1119	266.5	1235	291.5	1350	316.5	1466	341.5	1582
217.0	1005	242.0	1121	267.0	1237	292.0	1353	317.0	1468	342.0	1584
217.5	1008	242.5	1123	267.5	1239	292.5	1355	317.5	1471	342.5	1587
218.0	1010	243.0	1126	268.0	1242	293.0	1357	318.0	1473	343.0	1589
218.5	1012	243.5	1128	268.5	1244	293.5	1360	318.5	1475	343.5	1591
219.0	1015	244.0	1130	269.0	1246	294.0	1362	319.0	1478	344.0	1594
219.5	1017	244.5	1133	269.5	1248	294.5	1364	319.5	1480	344.5	1596
220.0	1019	245.0	1135	270.0	1251	295.0	1367	320.0	1482	345.0	1598
220.5	1021	245.5	1137	270.5	1253	295.5	1369	320.5	1485	345.5	1601
221.0	1024	246.0	1140	271.0	1255	296.0	1371	321.0	1487	346.0	1603
221.5	1026	246.5	1142	271.5	1258	296.5	1374	321.5	1489	346.5	1605
222.0	1028	247.0	1144	272.0	1260	297.0	1376	322.0	1492	347.0	1607
222.5	1031	247.5	1147	272.5	1262	297.5	1378	322.5	1494	347.5	1610
223.0	1033	248.0	1149	273.0	1265	298.0	1380	323.0	1496	348.0	1612
223.5	1035	248.5	1151	273.5	1267	298.5	1383	323.5	1499	348.5	1614
224.0	1038	249.0	1154	274.0	1269	299.0	1385	324.0	1501	349.0	1617
224.5	1040	249.5	1156	274.5	1272	299.5	1387	324.5	1503	349.5	1619

TABLE 1 *Continued*

Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F	Kin Vis, mm ² /s (cSt)	Equiv Say Univ Vis, SUS At 100°F
350.0	1621	375.0	1737	400.0	1853	425.0	1969	450.0	2085	475.0	2200
350.5	1624	375.5	1739	400.5	1855	425.5	1971	450.5	2087	475.5	2203
351.0	1626	376.0	1742	401.0	1858	426.0	1973	451.0	2089	476.0	2205
351.5	1628	376.5	1744	401.5	1860	426.5	1976	451.5	2092	476.5	2207
352.0	1631	377.0	1746	402.0	1862	427.0	1978	452.0	2094	477.0	2210
352.5	1633	377.5	1749	402.5	1865	427.5	1980	452.5	2096	477.5	2212
353.0	1635	378.0	1751	403.0	1867	428.0	1983	453.0	2098	478.0	2214
353.5	1638	378.5	1753	403.5	1869	428.5	1985	453.5	2101	478.5	2217
354.0	1640	379.0	1756	404.0	1872	429.0	1987	454.0	2103	479.0	2219
354.5	1642	379.5	1758	404.5	1874	429.5	1990	454.5	2105	479.5	2221
355.0	1645	380.0	1760	405.0	1876	430.0	1992	455.0	2108	480.0	2224
355.5	1647	380.5	1763	405.5	1878	430.5	1994	455.5	2110	480.5	2226
356.0	1649	381.0	1765	406.0	1881	431.0	1997	456.0	2112	481.0	2228
356.5	1651	381.5	1767	406.5	1883	431.5	1999	456.5	2115	481.5	2231
357.0	1654	382.0	1770	407.0	1885	432.0	2001	457.0	2117	482.0	2233
357.5	1656	382.5	1772	407.5	1888	432.5	2004	457.5	2119	482.5	2235
358.0	1658	383.0	1774	408.0	1890	433.0	2006	458.0	2122	483.0	2237
358.5	1661	383.5	1777	408.5	1892	433.5	2008	458.5	2124	483.5	2240
359.0	1663	384.0	1779	409.0	1895	434.0	2010	459.0	2126	484.0	2242
359.5	1665	384.5	1781	409.5	1897	434.5	2013	459.5	2129	484.5	2244
360.0	1668	385.0	1783	410.0	1899	435.0	2015	460.0	2131	485.0	2247
360.5	1670	385.5	1786	410.5	1902	435.5	2017	460.5	2133	485.5	2249
361.0	1672	386.0	1788	411.0	1904	436.0	2020	461.0	2136	486.0	2251
361.5	1675	386.5	1790	411.5	1906	436.5	2022	461.5	2138	486.5	2254
362.0	1677	387.0	1793	412.0	1909	437.0	2024	462.0	2140	487.0	2256
362.5	1679	387.5	1795	412.5	1911	437.5	2027	462.5	2142	487.5	2258
363.0	1682	388.0	1797	413.0	1913	438.0	2029	463.0	2145	488.0	2261
363.5	1684	388.5	1800	413.5	1916	438.5	2031	463.5	2147	488.5	2263
364.0	1686	389.0	1802	414.0	1918	439.0	2034	464.0	2149	489.0	2265
364.5	1689	389.5	1804	414.5	1920	439.5	2036	464.5	2152	489.5	2268
365.0	1691	390.0	1807	415.0	1922	440.0	2038	465.0	2154	490.0	2270
365.5	1693	390.5	1809	415.5	1925	440.5	2041	465.5	2156	490.5	2272
366.0	1695	391.0	1811	416.0	1927	441.0	2043	466.0	2159	491.0	2275
366.5	1698	391.5	1814	416.5	1929	441.5	2045	466.5	2161	491.5	2277
367.0	1700	392.0	1816	417.0	1932	442.0	2048	467.0	2163	492.0	2279
367.5	1702	392.5	1818	417.5	1934	442.5	2050	467.5	2166	492.5	2281
368.0	1705	393.0	1821	418.0	1936	443.0	2052	468.0	2168	493.0	2284
368.5	1707	393.5	1823	418.5	1939	443.5	2054	468.5	2170	493.5	2286
369.0	1709	394.0	1825	419.0	1941	444.0	2057	469.0	2173	494.0	2288
369.5	1712	394.5	1827	419.5	1943	444.5	2059	469.5	2175	494.5	2291
370.0	1714	395.0	1830	420.0	1946	445.0	2061	470.0	2177	495.0	2293
370.5	1716	395.5	1832	420.5	1948	445.5	2064	470.5	2180	495.5	2295
371.0	1719	396.0	1834	421.0	1950	446.0	2066	471.0	2182	496.0	2298
371.5	1721	396.5	1837	421.5	1953	446.5	2068	471.5	2184	496.5	2300
372.0	1723	397.0	1839	422.0	1955	447.0	2071	472.0	2187	497.0	2302
372.5	1726	397.5	1841	422.5	1957	447.5	2073	472.5	2189	497.5	2305
373.0	1728	398.0	1844	423.0	1960	448.0	2075	473.0	2191	498.0	2307
373.5	1730	398.5	1846	423.5	1962	448.5	2078	473.5	2193	498.5	2309
374.0	1733	399.0	1848	424.0	1964	449.0	2080	474.0	2196	499.0	2312
374.5	1735	399.5	1851	424.5	1966	449.5	2082	474.5	2198	499.5	2314
										500.0	2316

**TABLE 2 Conversion Factors, Kinematic Viscosity to Saybolt
Universal Viscosity**

Temperature, ° F	Conversion Factors	
	Factor A, 75 mm ² /s (cSt) and Under	Factor B, Over 75 mm ² /s (cSt)
0	0.994	4.604
10	0.995	4.607
20	0.995	4.610
30	0.996	4.613
40	0.996	4.615
50	0.997	4.618
60	0.998	4.621
70	0.998	4.624
80	0.999	4.627
90	0.999	4.630
100	1.000	4.632
110	1.001	4.635
120	1.001	4.638
130	1.002	4.641
140	1.002	4.644
150	1.003	4.647
160	1.004	4.649
170	1.004	4.652
180	1.005	4.655
190	1.005	4.658
200	1.006	4.661
210	1.007	4.664
220	1.007	4.666
230	1.008	4.669
240	1.009	4.672
250	1.009	4.675
260	1.010	4.678
270	1.010	4.680
280	1.011	4.683
290	1.012	4.686
300	1.012	4.689
310	1.013	4.692
320	1.013	4.695
330	1.014	4.697
340	1.015	4.700
350	1.015	4.703

**TABLE 3 Kinematic Viscosity to Saybolt Furol Viscosity
48 to 1300 mm²/s (cSt)**

Kin Vis, mm ² /s (cSt)	Equiv Say Furol Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furol Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furol Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furol Vis, SFS	
	At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F
48	25.1	...	...								
49	25.6	...	...								
50	26.0	25.2	100	48.6	48.4	150	71.5	72.1	200	94.8	96.0
51	26.5	25.6	101	49.1	48.9	151	72.0	72.6	201	95.2	96.5
52	27.0	26.1	102	49.5	49.3	152	72.4	73.1	202	95.7	96.9
53	27.4	26.5	103	50.0	49.8	153	72.9	73.5	203	96.2	97.4
54	27.9	27.0	104	50.4	50.3	154	73.4	74.0	204	96.6	97.9
55	28.3	27.4	105	50.9	50.7	155	73.8	74.5	205	97.1	98.4
56	28.8	27.9	106	51.3	51.2	156	74.3	75.0	206	97.6	98.8
57	29.2	28.4	107	51.8	51.7	157	74.8	75.4	207	98.0	99.8
58	29.7	28.8	108	52.3	52.2	158	75.2	75.9	208	98.5	99.8
59	30.1	29.3	109	52.7	52.6	159	75.7	76.4	209	99.0	100.3
60	30.6	29.7	110	53.2	53.1	160	76.1	76.9	210	99.4	100.8
61	31.1	30.2	111	53.6	53.6	161	76.6	77.4	211	99.9	101.2
62	31.5	30.6	112	54.1	54.1	162	77.1	77.8	212	100.4	101.7
63	32.0	31.1	113	54.5	54.5	163	77.5	78.3	213	100.9	102.2
64	32.4	31.6	114	55.0	55.0	164	78.0	78.8	214	101.3	102.7
65	32.9	32.0	115	55.4	55.5	165	78.5	79.3	215	101.8	103.1
66	33.3	32.5	116	55.9	55.9	166	78.9	79.7	216	102.3	103.6
67	33.8	33.0	117	56.3	56.4	167	79.4	80.2	217	102.7	104.1
68	34.2	33.4	118	56.8	56.9	168	79.9	80.7	218	103.2	104.6
69	34.7	33.9	119	57.3	57.4	169	80.3	81.2	219	103.7	105.1
70	35.1	34.3	120	57.7	57.8	170	80.8	81.6	220	104.1	105.5
71	35.6	34.8	121	58.2	58.3	171	81.2	82.1	221	104.6	106.0
72	36.0	35.3	122	58.6	58.8	172	81.7	82.6	222	105.1	106.5
73	36.5	35.7	123	59.1	59.3	173	82.2	83.1	223	105.6	107.0
74	36.9	36.2	124	59.5	59.7	174	82.6	83.6	224	106.0	107.4
75	37.4	36.7	125	60.0	60.2	175	83.1	84.0	225	106.5	107.9
76	37.8	37.1	126	60.5	60.7	176	83.6	84.5	226	106.9	108.4
77	38.3	37.6	127	60.9	61.2	177	84.0	85.0	227	107.4	108.9
78	38.7	38.1	128	61.4	61.6	178	84.5	85.5	228	107.9	109.4
79	39.2	38.5	129	61.8	62.1	179	85.0	85.9	229	108.3	109.8
80	39.6	39.0	130	62.3	62.6	180	85.4	86.4	230	108.8	110.3
81	40.1	39.5	131	62.8	63.1	181	85.9	86.9	231	109.3	110.8
82	40.5	39.9	132	63.2	63.5	182	86.4	87.4	232	109.8	111.3
83	41.0	40.4	133	63.7	64.0	183	86.8	87.9	233	110.2	111.8
84	41.4	40.9	134	64.1	64.5	184	87.3	88.3	234	110.7	112.2
85	41.9	41.3	135	64.6	65.0	185	87.8	88.8	235	111.2	112.7
86	42.3	41.8	136	65.1	65.4	186	88.2	89.3	236	111.6	113.2
87	42.8	42.3	137	65.5	65.9	187	88.7	89.8	237	112.1	113.7
88	43.2	42.7	138	66.0	66.4	188	89.2	90.2	238	112.6	114.1
89	43.7	43.2	139	66.4	66.9	189	89.6	90.7	239	113.0	114.6
90	44.1	43.7	140	66.9	67.3	190	90.1	91.2	240	113.5	115.1
91	44.6	44.1	141	67.4	67.8	191	90.6	91.7	241	114.0	115.6
92	45.0	44.6	142	67.8	68.3	192	91.0	92.2	242	114.4	116.1
93	45.5	45.1	143	68.3	68.8	193	91.5	92.6	243	114.9	116.5
94	45.9	45.6	144	68.7	69.3	194	92.0	93.1	244	115.4	117.0
95	46.4	46.0	145	69.2	69.7	195	92.4	93.6	245	115.9	117.5
96	46.8	46.5	146	69.7	70.2	196	92.9	94.1	246	116.3	118.0
97	47.3	47.0	147	70.1	70.7	197	93.4	94.5	247	116.8	118.5
98	47.7	47.4	148	70.6	71.2	198	93.8	95.0	248	117.3	118.9
99	48.2	47.9	149	71.0	71.6	199	94.3	95.5	249	117.7	119.4

TABLE 3 *Continued*

Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS	
	At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F
250	118.2	119.9	300	141.7	143.8	350	165.2	167.8	400	188.8	191.7
251	118.7	120.4	301	142.2	144.3	351	165.7	168.2	401	189.3	192.2
252	119.1	120.8	302	142.6	144.8	352	166.2	168.7	402	189.7	192.7
253	119.6	121.3	303	143.1	145.3	353	166.6	169.2	403	190.2	193.2
254	120.1	121.8	304	143.6	145.7	354	167.1	169.7	404	190.7	193.6
255	120.5	122.3	305	144.0	146.2	355	167.6	170.2	405	191.1	194.1
256	121.0	122.8	306	144.5	146.7	356	168.1	170.6	406	191.6	194.6
257	121.5	123.2	307	145.0	147.2	357	168.5	171.1	407	192.1	195.1
258	122.0	123.7	308	145.5	147.7	358	169.0	171.6	408	192.6	195.5
259	122.4	124.2	309	145.9	148.1	359	169.5	172.1	409	193.0	196.0
260	122.9	124.7	310	146.4	148.6	360	169.9	172.6	410	193.5	196.5
261	123.4	125.2	311	146.9	149.1	361	170.4	173.0	411	194.0	197.0
262	123.8	125.6	312	147.3	149.6	362	170.9	173.5	412	194.4	197.5
263	124.3	126.1	313	147.8	150.0	363	171.4	174.0	413	194.9	197.9
264	124.8	126.6	314	148.3	150.5	364	171.8	174.5	414	195.4	198.4
265	125.2	127.1	315	148.8	151.0	365	172.3	174.9	415	195.8	198.9
266	125.7	127.5	316	149.2	151.5	366	172.8	175.4	416	196.3	199.4
267	126.2	128.0	317	149.7	152.0	367	173.2	175.9	417	196.8	199.9
268	126.7	128.5	318	150.2	152.4	368	173.7	176.4	418	197.3	200
269	127.1	129.0	319	150.6	152.9	369	174.2	176.9	419	197.7	201
270	127.6	129.5	320	151.1	153.4	370	174.6	177.3	420	198.2	201
271	128.1	129.9	321	151.6	153.9	371	175.1	177.8	421	198.7	202
272	128.5	130.4	322	152.0	154.4	372	175.6	178.3	422	199.1	202
273	129.0	130.9	323	152.5	154.8	373	176.1	178.8	423	199.6	203
274	129.5	131.4	324	153.0	155.3	374	176.5	179.3	424	200	203
275	129.9	131.9	325	153.5	155.8	375	177.0	179.7	425	201	204
276	130.4	132.3	326	153.9	156.3	376	177.5	180.2	426	201	204
277	130.9	132.8	327	154.4	156.7	377	177.9	180.7	427	202	205
278	131.4	133.3	328	154.9	157.2	378	178.4	181.2	428	202	205
279	131.8	133.8	329	155.3	157.7	379	178.9	181.7	429	202	206
280	132.3	134.2	330	155.8	158.2	380	179.4	182.1	430	203	206
281	132.8	134.7	331	156.3	158.7	381	179.8	182.6	431	203	207
282	133.2	135.2	332	156.8	159.1	382	180.3	183.1	432	204	207
283	133.7	135.7	333	157.2	159.6	383	180.8	183.6	433	204	208
284	134.2	136.2	334	157.7	160.1	384	181.2	184.1	434	205	208
285	134.6	136.6	335	158.2	160.6	385	181.7	184.5	435	205	208
286	135.1	137.1	336	158.6	161.1	386	182.2	185.0	436	206	209
287	135.6	137.6	337	159.1	161.5	387	182.7	185.5	437	206	209
288	136.1	138.1	338	159.6	162.0	388	183.1	186.0	438	207	210
289	136.5	138.6	339	160.0	162.5	389	183.6	186.4	439	207	210
290	137.0	139.0	340	160.5	163.0	390	184.1	186.9	440	208	211
291	137.5	139.5	341	161.0	163.5	391	184.5	187.4	441	208	211
292	137.9	140.0	342	161.5	163.9	392	185.0	187.9	442	209	212
293	138.4	140.5	343	161.9	164.4	393	185.5	188.4	443	209	212
294	138.9	140.9	344	162.4	164.9	394	186.0	188.8	444	210	213
295	139.3	141.4	345	162.9	165.4	395	186.4	189.3	445	210	213
296	139.8	141.9	346	163.3	165.8	396	186.9	189.8	446	210	214
297	140.3	142.4	347	163.8	166.3	397	187.4	190.3	447	211	214
298	140.8	142.9	348	164.3	166.8	398	187.8	190.8	448	211	215
299	141.2	143.3	349	164.8	167.3	399	188.3	191.2	449	212	215

TABLE 3 *Continued*

Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS	
	At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F
450	212	216	500	236	240	550	259	264	600	283	288
451	213	216	501	236	240	551	260	264	601	284	288
452	213	217	502	237	241	552	260	265	602	284	288
453	214	217	503	237	241	553	261	265	603	284	289
454	214	218	504	238	242	554	261	265	604	285	289
455	215	218	505	238	242	555	262	266	605	285	290
456	215	219	506	239	242	556	262	266	606	286	290
457	216	219	507	239	243	557	263	267	607	286	291
458	216	220	508	240	243	558	263	267	608	287	291
459	217	220	509	240	244	559	264	268	609	287	292
460	217	220	510	241	244	560	264	268	610	288	292
461	218	221	511	241	245	561	265	269	611	288	293
462	218	221	512	242	245	562	265	269	612	289	293
463	218	222	513	242	246	563	266	270	613	289	294
464	219	222	514	243	246	564	266	270	614	290	294
465	219	223	515	243	247	565	267	271	615	290	295
466	220	223	516	243	247	566	267	271	616	291	295
467	220	224	517	244	248	567	268	272	617	291	296
468	221	224	518	244	248	568	268	272	618	292	296
469	221	225	519	245	249	569	268	273	619	292	297
470	222	225	520	245	249	570	269	273	620	292	297
471	222	226	521	246	250	571	269	274	621	293	298
472	223	226	522	246	250	572	270	274	622	293	298
473	223	227	523	247	251	573	270	275	623	294	299
474	224	227	524	247	251	574	271	275	624	294	299
475	224	228	525	248	252	575	271	276	625	295	300
476	225	228	526	248	252	576	272	276	626	295	300
477	225	229	527	249	253	577	272	277	627	296	300
478	226	229	528	249	253	578	273	277	628	296	301
479	226	230	529	250	254	579	273	277	629	297	301
480	226	230	530	250	254	580	274	278	630	297	302
481	227	231	531	251	254	581	274	278	631	298	302
482	227	231	532	251	255	582	275	279	632	298	303
483	228	231	533	251	255	583	275	279	633	299	303
484	228	232	534	252	256	584	276	280	634	299	304
485	229	232	535	252	256	585	276	280	635	300	304
486	229	233	536	253	257	586	276	281	636	300	305
487	230	233	537	253	257	587	277	281	637	301	305
488	230	234	538	254	258	588	277	282	638	301	306
489	231	234	539	254	258	589	278	282	639	301	306
490	231	235	540	255	259	590	278	283	640	302	307
491	323	235	541	255	259	591	279	283	641	302	307
492	232	236	542	256	260	592	279	284	642	303	308
493	233	236	543	256	260	593	280	284	643	303	308
494	233	237	544	257	261	594	280	285	644	304	309
495	234	237	545	257	261	595	281	285	645	304	309
496	234	238	546	258	262	596	281	286	646	305	310
497	234	238	547	258	262	597	282	286	647	305	310
498	235	239	548	259	263	598	282	287	648	306	311
499	235	239	549	259	263	599	283	287	649	306	311

TABLE 3 *Continued*

Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS	
	At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F
650	307	311	700	330	335	750	354	359	800	377	383
651	307	312	701	331	336	751	354	360	802	378	384
652	308	312	702	331	336	752	355	360	804	379	385
653	308	313	703	332	337	753	355	361	806	380	386
654	309	313	704	332	337	754	356	361	808	381	387
655	309	314	705	333	338	755	356	362	810	382	388
656	309	314	706	333	338	756	357	362	812	383	389
657	310	315	707	334	339	757	357	363	814	384	390
658	310	315	708	334	339	758	358	363	816	385	391
659	311	316	709	334	340	759	358	364	818	386	392
660	311	316	710	335	340	760	359	364	820	387	393
661	312	317	711	335	341	761	359	365	822	388	394
662	312	317	712	336	341	762	359	365	824	389	395
663	313	318	713	336	342	763	360	366	826	390	396
664	313	318	714	337	342	764	360	366	828	391	397
665	314	319	715	337	343	765	361	367	830	392	398
666	314	319	716	338	343	766	361	367	832	392	399
667	315	320	717	338	344	767	362	368	834	393	400
668	315	320	718	339	344	768	362	368	836	394	401
669	316	321	719	339	345	769	363	369	838	395	402
670	316	321	720	340	345	770	363	369	840	396	403
671	317	322	721	340	346	771	364	369	842	397	403
672	317	322	722	341	346	772	364	370	844	398	404
673	317	323	723	341	346	773	365	370	846	399	405
674	318	323	724	342	347	774	365	371	848	400	406
675	318	323	725	342	347	775	366	371	850	401	407
676	319	324	726	342	348	776	366	372	852	402	408
677	319	324	727	343	348	777	367	372	854	403	409
678	320	325	728	343	349	778	367	373	856	404	410
679	320	325	729	344	349	779	367	373	858	405	411
680	321	326	730	344	350	780	368	374	860	406	412
681	321	326	731	345	350	781	368	374	862	407	413
682	322	327	732	345	351	782	369	375	864	408	414
683	322	327	733	346	351	783	369	375	866	409	415
684	323	328	734	346	352	784	370	376	868	409	416
685	323	328	735	347	352	785	370	376	870	410	417
686	324	329	736	347	353	786	371	377	872	411	418
687	324	329	737	348	353	787	371	377	874	412	419
688	325	330	738	348	354	788	372	378	876	413	420
689	325	330	739	349	354	789	372	378	878	414	421
690	326	331	740	349	355	790	373	379	880	415	422
691	326	331	741	350	355	791	373	379	882	416	423
692	326	332	742	350	356	792	374	380	884	417	424
693	327	332	743	351	356	793	374	380	886	418	425
694	327	333	744	351	357	794	375	380	888	419	426
695	328	333	745	351	357	795	375	381	890	420	426
696	328	334	746	352	357	796	375	381	892	421	427
697	329	334	747	352	358	797	376	382	894	422	428
698	329	334	748	353	358	798	376	382	896	423	429
699	330	335	749	353	359	799	377	383	898	424	430

TABLE 3 *Continued*

Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		Kin Vis, mm ² /s (cSt)	Equiv Say Furoil Vis, SFS		
	At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F		At 122°F	At 210°F	
900	425	431	1000	472	479	1100	519	527	1200	566	575	
902	425	432	1002	473	480	1102	520	528	1202	567	576	
904	426	433	1004	474	481	1104	521	529	1204	568	577	
906	427	434	1006	475	482	1106	522	530	1206	569	578	
908	428	435	1008	475	483	1108	523	531	1208	570	579	
910	429	436	1010	476	484	1110	524	532	1210	571	580	
912	430	437	1012	477	485	1112	525	533	1212	572	581	
914	431	438	1014	478	486	1114	525	534	1214	573	582	
916	432	439	1016	479	487	1116	526	535	1216	574	583	
918	433	440	1018	480	488	1118	527	536	1218	575	584	
920	434	441	1020	481	489	1120	528	537	1220	575	585	
922	435	442	1022	482	490	1122	529	538	1222	576	586	
924	436	443	1024	483	491	1124	530	539	1224	577	587	
926	437	444	1026	484	492	1126	531	540	1226	578	588	
928	438	445	1028	485	493	1128	532	541	1228	579	588	
930	439	446	1030	486	494	1130	533	542	1230	580	589	
932	440	447	1032	487	495	1132	534	542	1232	581	590	
934	441	448	1034	488	495	1134	535	543	1234	582	591	
936	442	449	1036	489	496	1136	536	544	1236	583	592	
938	442	449	1038	490	497	1138	537	545	1238	584	593	
940	443	450	1040	491	498	1140	538	546	1240	585	594	
942	444	451	1042	492	499	1142	539	547	1242	586	595	
944	445	452	1044	492	500	1144	540	548	1244	587	596	
946	446	453	1046	493	501	1146	541	549	1246	588	597	
948	447	454	1048	494	502	1148	542	550	1248	589	598	
950	448	455	1050	495	503	1150	542	551	1250	590	599	
952	449	456	1052	496	504	1152	543	552	1252	591	600	
954	450	457	1054	497	505	1154	544	553	1254	592	601	
956	451	458	1056	498	506	1156	545	554	1256	592	602	
958	452	459	1058	499	507	1158	546	555	1258	593	603	
960	453	460	1060	500	508	1160	547	556	1260	594	604	
962	454	461	1062	501	509	1162	548	557	1262	595	605	
964	455	462	1064	502	510	1164	549	558	1264	596	606	
966	456	463	1066	503	511	1166	550	559	1266	597	607	
968	457	464	1068	504	512	1168	551	560	1268	598	608	
970	458	465	1070	505	513	1170	552	561	1270	599	609	
972	459	466	1072	506	514	1172	553	562	1272	600	610	
974	459	467	1074	507	515	1174	554	563	1274	601	611	
976	460	468	1076	508	516	1176	555	564	1276	602	611	
978	461	469	1078	509	517	1178	556	565	1278	603	612	
980	462	470	1080	509	518	1180	557	565	1280	604	613	
982	463	471	1082	510	518	1182	558	566	1282	605	614	
984	464	472	1084	511	519	1184	559	567	1284	606	615	
986	465	472	1086	512	520	1186	559	568	1286	607	616	
988	466	473	1088	513	521	1188	560	569	1288	608	617	
990	467	474	1090	514	522	1190	561	570	1290	609	618	
992	468	475	1092	515	523	1192	562	571	1292	609	619	
994	469	476	1094	516	524	1194	563	572	1294	610	620	
996	470	477	1096	517	525	1196	564	573	1296	611	621	
998	471	478	1098	518	526	1198	565	574	1298	612	622	
										1300	613	623

APPENDIX

(Nonmandatory Information)

X1. EXAMPLES ILLUSTRATING VISCOSITY CONVERSIONS

X1.1 Example 1

X1.1.1 What is the Saybolt Universal viscosity equivalent to a kinematic viscosity of $74.5 \text{ mm}^2/\text{s}$ (cSt) at 100°F ?

X1.1.2 Enter **Table 1** with the kinematic viscosity of $74.5 \text{ mm}^2/\text{s}$ (cSt) and note that at a temperature of 100°F the equivalent Saybolt Universal viscosity is 346 SUS.

X1.2 Example 2

X1.2.1 What is the Saybolt Universal viscosity equivalent to a kinematic viscosity of $24.87 \text{ mm}^2/\text{s}$ (cSt) at 100°F ?

X1.2.2 Enter **Table 1** with $24.85 \text{ mm}^2/\text{s}$ (cSt) and note that the equivalent Saybolt Universal viscosity at 100°F is 118.7 SUS. Likewise, enter the table with $24.90 \text{ mm}^2/\text{s}$ (cSt) and find that the equivalent Saybolt Universal viscosity at 100°F is 118.9 SUS. An increase of $0.05 \text{ mm}^2/\text{s}$ (cSt) is equal to a corresponding increase of 0.2 SUS. Therefore, by simple proportion, an increase of $0.02 \text{ mm}^2/\text{s}$ (cSt) kinematic viscosity increases the equivalent Saybolt Universal viscosity by $0.02 \div 0.05 \times 0.2 = 0.08$ SUS. Hence, the Saybolt Universal viscosity at 100°F equivalent to $24.87 \text{ mm}^2/\text{s}$ (cSt) at 100°F is $118.7 + 0.08 = 118.78$ and round off to 118.8 SUS.

X1.3 Example 3

X1.3.1 What is the Saybolt Universal viscosity equivalent to a kinematic viscosity of $745 \text{ mm}^2/\text{s}$ (cSt) at 100°F ?

X1.3.2 Multiply 745 by 4.632, the *B* factor at 100, to obtain 3451 SUS.

X1.4 Example 4

X1.4.1 What is the Saybolt Universal viscosity equivalent to a kinematic viscosity of $54.4 \text{ mm}^2/\text{s}$ (cSt) at 180°F ?

X1.4.2 From **Table 1** convert the kinematic viscosity at 180°F of 54.4 to the equivalent Saybolt Universal viscosity at 100°F , namely 253 SUS. From **Table 2** obtain Factor *A* for conversion at a temperature 180°F 1.005. Multiply 253 SUS by 1.005 to obtain 254 SUS.

X1.5 Example 5

X1.5.1 What is the Saybolt Universal viscosity equivalent to a kinematic viscosity of $89.95 \text{ mm}^2/\text{s}$ (cSt) at 40°F ?

X1.5.2 From **Table 2** obtain Factor *B* for a temperature of $40^\circ\text{F} = 4.615$. Multiply 89.95 by 4.615 to obtain 415 SUS.

X1.6 Example 6

X1.6.1 What is the Saybolt Furol viscosity equivalent to a kinematic viscosity of $231 \text{ mm}^2/\text{s}$ (cSt) at 122°F ?

X1.6.2 Enter **Table 3** with $231 \text{ mm}^2/\text{s}$ (cSt) and note that the equivalent Saybolt Furol viscosity at 122°F is 109.3 SFS.

X1.7 Example 7

X1.7.1 What is the Saybolt Furol viscosity at 210°F equivalent to a kinematic viscosity of $287 \text{ mm}^2/\text{s}$ (cSt) at 210°F ?

X1.7.2 Enter **Table 3** with $287 \text{ mm}^2/\text{s}$ (cSt) and note that the equivalent Saybolt Furol viscosity at 210°F is 137.6 SFS.

X1.8 Example 8

X1.8.1 What is the Saybolt Furol viscosity equivalent to a kinematic viscosity of $276.2 \text{ mm}^2/\text{s}$ (cSt) at 122°F ?

X1.8.2 Enter **Table 3** with $276 \text{ mm}^2/\text{s}$ (cSt) and note that the equivalent Saybolt Furol viscosity at 122°F is 130.4 SFS. Likewise, enter the table with $277 \text{ mm}^2/\text{s}$ (cSt) and note that the equivalent Saybolt Furol viscosity is 130.9 SFS. An increase of $1.0 \text{ mm}^2/\text{s}$ (cSt) is equal to a corresponding increase of 0.5 SFS. Therefore, by simple proportion, an increase of $0.2 \text{ mm}^2/\text{s}$ (cSt) increased the Saybolt Furol viscosity equivalent of $276 \text{ mm}^2/\text{s}$ (cSt) by $0.2/0.1 \times 0.5 = 0.1$ SFS. Hence, the Saybolt Furol viscosity at 122°F equivalent to $276.2 \text{ mm}^2/\text{s}$ (cSt) kinematic viscosity at 122°F is $130.4 + 0.1 = 130.5$ SFS.

X1.9 Example 9

X1.9.1 What is the Saybolt Furol viscosity equivalent to $1500 \text{ mm}^2/\text{s}$ (cSt) at 122°F ?

X1.9.2 Applying Eq 3: $1500 \times 0.4717 = 707.55$ and rounds off to 708 SFS.

SUMMARY OF CHANGES

Subcommittee D02.07 has identified the location of selected changes to this standard since the last issue (D 2161 – 04) that may impact the use of this standard. (Approved April 1, 2005.)

- (1) Changed the primary unit of kinematic viscosity from cSt to mm²/s.
- (2) Changed the secondary unit of kinematic viscosity from mm²/s to cSt.
- (3) Corrected spelling and typographical errors.

Subcommittee D02.07 has identified the location of selected changes to this standard since the last issue (D 2161 – 93 (1999)^{e2}) that may impact the use of this standard. (Approved Dec. 1, 2004.)

- (1) Changed the viscosity of water at 20°C from 1.0038 to 1.0034 mm²/s.

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